Post-Capitalist Labs Imprint · Working Draft Monograph

DISTRIBUTED PARTICIPATORY ECONOMICS PROTOCOL

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons



DESIGNED FOR THREE INTERDISCIPLINARY AUDIENCES:

- Advocates & Cooperative Practitioners: Council democracy, balanced job complexes, and historical lessons from Catalonia (1936), the Paris Commune (1871), and Classical Athens.
- Economists & Mechanism Designers: Formal welfare theorems, Sandström's double-entry accounting, pollution demand revelation (CAPs), and non-convex ADMM convergence.
- Systems Programmers & Cryptographers: Local-first CRDTs, BLAKE3 Sparse Merkle-Sum Trees, deterministic WASM IFB engines, and NIST FIPS 203/204/205 post-quantum security.

POST-CAPITALIST LABS IMPRINT · https://github.com/Post-Capitalist-Labs Working Draft · Nov 2026

Leveraging technology to amplify social movements · Decentralised consensus, mechanism design & democratic coordination

Post-Capitalist Labs Imprint · Open-Source Research Monograph Series

Distributed Participatory Economics Protocol (DPEP)

A Rigorous, Source-Grounded Feasibility Study and Post-Quantum Technical Architecture for Democratic Planning, Self-Management, and the Productive Commons

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About Post-Capitalist Labs (PCL)

At Post-Capitalist Labs, we harness the transformative power of technology to bolster social movements and build directly democratic alternatives to capitalism. Our work spans decentralised consensus, applied cryptography, operations research, and democratic mechanism design.

Our Vision: We envisage a world where solidarity, care, classlessness, diversity, self-reflection, and direct democracy are not just ideals, but the foundations of everyday life. Our goal is to make this vision a reality through the thoughtful application of open-source technology, where people remain in control.

Our Approach:

- Learning from History: Analyzing past and present democratic movements and self-managed collectives to guide institutional design.
- Emancipatory Modeling: Developing rigorous mathematical and computational models for equitable, non-market resource coordination.
- Open-Source Engineering: Turning theoretical models into verifiable, accessible, community-driven proofs of concept and protocols.

Organization Profile: https://github.com/Post-Capitalist-Labs
Repository: https://github.com/Post-Capitalist-Labs/distributed-participatory-economics-protocol

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(1981), Mansbridge (1980), and Ostrom (1990). Tier 4 (Code Audited, JVM Not Re-Executed): pequod-cljs was audited via Python scripts without re-running Leiningen. Short quotations and epigraphs are reproduced verbatim with attribution under fair dealing / fair use.

Interdisciplinary Navigation Guide

How to Read This Book Across Disciplines

This monograph bridges libertarian-socialist institutional theory, formal welfare and planning economics, and modern distributed systems cryptography. Depending on your primary background, we recommend the following reading paths:

FOR ADVOCATES, COOPERATORS & ORGANIZERS — Institutional Self-Management Without Hierarchy

Start with Chapter 1 (Executive Summary) and Chapter 2 (Historical & Institutional Inspirations) to see how lessons from the 1936–1939 Spanish Revolution, the 1871 Paris Commune, Classical Athens, Occupy Wall Street, Mondragón, Porto Alegre Participatory Budgeting, Rojava, and Elinor Ostrom's commons principles are translated into concrete protocol safeguards against bureaucratic capture, meeting fatigue, and inter-regional inequality. Then read Chapter 3 for the complete traceability matrix linking Michael Albert and Robin Hahnel's 1991 and 2021 books to our design.

FOR ECONOMISTS & OPERATIONS RESEARCHERS — Formal Models, Accounting & Non-Convex Convergence

Focus on Chapter 4 (Formal Economic Model), which unites Albert and Hahnel's Formal Model of a Participatory Economy (FMPE) with Anders Sandström's double-entry Society Account and Robin Hahnel's Pollution Demand-Revealing Mechanism (PDRM). See Section 4.4 for our source-code audit of pequod-cljs and Algorithm 3 (Safeguarded Primal-Dual / ADMM Proximal Regularisation), alongside Chapter 8 and Chapter 10 for our synthesis of John O'Neill's (2024) ecological incommensurability critique and non-convex convergence experiments.

FOR SYSTEMS PROGRAMMERS & CRYPTOGRAPHERS — Local-First P2P, Merkle-Sum Trees & Post-Quantum Security

Jump directly to Chapter 4 (Section 4.5 State Machine & Protobuf Schemas), Chapter 5 (Four-Layer Architecture D & 6-Role Mobile Taxonomy), Chapter 6 (Weighted Stack Comparison), Chapter 7 (NIST FIPS 203/204/205 Post-Quantum Cryptography, Non-Plutocratic Sybil Resistance & Zero-Knowledge Budget Proofs), and Chapter 9 (Implementable Proof-of-Concept Specification & Multi-Scale Benchmark Programme).

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List of Abbreviations and Mathematical Notation

Because this monograph integrates participatory economics, labor and cooperative history, operations research, distributed systems, and post-quantum cryptography, this reference table defines every abbreviation and mathematical symbol used across Chapters 1–10.

A. Participatory Economics, Accounting, and Bibliographic Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
BJC	Balanced Job Complex	Bundle of workplace tasks balanced for empowerment and desirability within and across workplaces (PEPE Ch. 2).
CAP	Community of Affected Parties	Ecological/geographic collective of residents impacted by a specific pollutant in an airshed or watershed (DEP Ch. 7).
CC (CC_h)	Consumers' Council	Democratic neighbourhood consumer council or nested ward/city/regional consumer federation.
CFB	Consumption Facilitation Board	Informational board assisting consumer councils and federations in evaluating collective consumption projects (LF Ch. 6).
CLUEA	Consejo Levantino Unificado de Exportación Agrícola	Unified Levantine Council for Agricultural Export organised by the CNT and UGT in Valencia (1936–1939).
CNT	Confederación Nacional del Trabajo	Anarcho-syndicalist National Confederation of Labour in the 1936–1939 Spanish Revolution.
CPR	Common-Pool Resource	Shared natural or human-made resource system governed by collective-choice rules (Ostrom 1990).
DEP	Democratic Economic Planning	Primary reference volume by Robin Hahnel et al. (Routledge, 2021).
DPEP	Distributed Participatory Economics Protocol	The open-source decentralised coordination protocol specified in this monograph.
EFB	Employment Facilitation Board	Facilitation board assisting workers transferring between councils and balancing cross-workplace BJCs (LF Ch. 6).
EJPE	Erasmus Journal for Philosophy and Economics	Academic journal hosting the Winter 2024 symposium on Hahnel's Democratic Economic Planning.
FMPE	Formal Model of a Participatory Economy	Mathematical formalisation of participatory planning analyzed in PEPE Chapter 5 (pp. 73–106).
FTE	Full-Time Equivalent	Unit of engineering or research staffing effort across roadmap scenarios.
GA	General Assembly	Direct plenary deliberative assembly of a workplace, village collective, or horizontal movement.

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Abbreviation	Full Term	Definition & Role in Monograph
GDP	Gross Domestic Product	Aggregate social value of net final output (Y^{GDP}) cleared in the Society Account.
GHG	Greenhouse Gas	Atmospheric emissions (CO_2-equivalent) subject to hard biophysical ceilings ($E_z^{\max}$).
HFB	Household Facilitation Board	Facilitation board assisting individuals and families with housing and neighbourhood membership (LF Ch. 6).
IF (IF_k)	Industry Federation	Federation of worker councils within industry k managing sector capacity, R&D, and SB/SC appeals.
IFB	Iteration Facilitation Board	Informational board/engine that aggregates supply and demand and updates indicative prices via public rules.
LF	Looking Forward	Primary reference volume by Michael Albert and Robin Hahnel (South End Press, 1991).
Parecon / PE	Participatory Economics	Post-capitalist economic model developed by Michael Albert and Robin Hahnel.
PB	Participatory Budgeting / Planning Bureau	Municipal democratic budgeting (Porto Alegre, 1989); also denotes the abstract Planning Bureau in FMPE Ch. 5.
PCL	Post-Capitalist Labs	Open-source research and engineering collective publishing this monograph (https://github.com/Post-Capitalist-Labs).
PDRM	Pollution Demand-Revealing Mechanism	Iterative procedure pricing local pollutant damages via CAP emission tolerances and credits (DEP Ch. 7).
PEP	Participatory Economy Project	Research collective maintaining participatoryeconomy.org and the 2023 User Story Mapping.
PEPE	The Political Economy of Participatory Economics	Primary technical reference volume by Michael Albert and Robin Hahnel (Princeton University Press, 1991).
PLAC	Parecon: Life After Capitalism	Comprehensive institutional and daily-life synthesis of Participatory Economics by Michael Albert (Verso Books, 2003).
PFB	Production Facilitation Board	Board assisting worker councils with production externalities, scale economies, and technology choice (LF Ch. 6).
PIPA	Practically Identical Production Activities	Equivalence classes $\{PIPA\}_i$ partitioning workplace tasks by qualitative empowerment effect (PEPE Ch. 2, p. 26).
SB / SC	Social Benefit / Social Cost	Ratio $SB_j(P)/SC_j(P)$ measuring the social value of a worker council's outputs relative to the opportunity cost of its inputs and endowments (≥ 1 constraint).
SKU	Stock Keeping Unit	Granular product variant (size, color, style) resolved at intra-year fulfilment from coarse planning categories.

Continued from previous page

Abbreviation	Full Term	Definition & Role in Monograph
UFB	Updating Facilitation Board	Facilitation board mediating intra-year plan adjustments when demand shifts or supply shocks exceed slack (LF Ch. 6).
UGT	Unión General de Trabajadores	Socialist General Union of Workers that co-organised industrial and agrarian collectives in Spain (1936–1939).
WC (WC_j)	Workers' Council	Democratic workplace collective where each worker has one vote and holds a Balanced Job Complex.

B. Operations Research, AI, Distributed Systems, and Cryptography Abbreviations

Abbreviation	Full Term	Definition & Role in Monograph
ADMM	Alternating Direction Method of Multipliers	Augmented-Lagrangian primal-dual optimisation method with proximal damping used in Algorithm 3.
AP / CP	Availability / Consistency under Partition	CAP theorem properties: local-first drafting is AP (available offline), while Epoch BFT price finality is CP.
ARIMA / ETS	AutoRegressive Integrated Moving Average / Error-Trend-Seasonality	Classical statistical time-series forecasting models used for baseline demand and SKU disaggregation.
BFT	Byzantine Fault Tolerance	Distributed consensus property guaranteeing safety even when up to $f < n/3$ validator nodes act maliciously.
BLAKE3	BLAKE3 Cryptographic Hash	High-speed 256-bit cryptographic hash function used for Sparse Merkle-Sum Trees and content addressing.
BSP	Bulk-Synchronous Parallel	Execution model alternating parallel local computation rounds with synchronised global aggregation checkpoints.
CBOR (CDE)	Concise Binary Object Representation	RFC 8949 Canonical Deterministic Encoding used for all signed DPEP messages with fixed 10^{-6} micro-unit scale.
CID	Content Identifier	Self-certifying cryptographic hash address referencing a redactable qualitative List or plan blob.
DA	Data Availability	Verification that all leaf proposals of a round root are downloadable by auditing nodes (60 MB compressed/round).
FLP	Fully Linear Proof	Information-theoretic zero-knowledge proof over secret shares used in Prio3 VDAFs (Boneh et al. 2019).
Prio3 / VDAF	Verifiable Distributed Aggregation Function	IETF CFRG draft-irtf-cfrg-vdaf-16 protocol binding secret-shared household vectors to budget compliance proofs.
CRDT	Conflict-Free Replicated Data Type	Data structure (Loro / Automerge) allowing offline concurrent edits in council workspaces that merge deterministically.

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Abbreviation	Full Term	Definition & Role in Monograph
CRQC	Cryptographically Relevant Quantum Computer	Future fault-tolerant quantum computer capable of breaking classical elliptic-curve and RSA cryptography via Shor's algorithm.
DAO	Decentralised Autonomous Organisation	Blockchain governance construct critiqued in Chapter 2 for conflating token voting with council democracy.
DTN	Delay-Tolerant Networking	Store-and-forward networking (including LoRa mesh and USB data-mules) for intermittently connected regions.
EEIO / LCA	Environmentally Extended Input-Output / Lifecycle Assessment	Matrix accounting methods estimating cradle-to-gate resource and emission footprints across supply chains.
FIPS	Federal Information Processing Standards	NIST post-quantum cryptographic standards finalised in August 2024: FIPS 203, FIPS 204, and FIPS 205.
FRI / STARK	Fast Reed-Solomon Interactive Oracle Proof / Scalable Transparent ARgument of Knowledge	Post-quantum, hash-based zero-knowledge proof system (Plonky3) requiring no trusted setup.
IEEE-754	IEEE Floating-Point Standard	Hardware floating-point representation replaced in DPEP core rules by deterministic 64-bit fixed-point integers.
KKT	Karush-Kuhn-Tucker Conditions	First-order necessary and sufficient optimality conditions for constrained nonlinear/conic optimisation.
LP / QP / MILP	Linear / Quadratic / Mixed-Integer Linear Programming	Mathematical optimisation paradigms solved locally via Clarabel.rs and HiGHS.
MAPE	Mean Absolute Percentage Error	Accuracy metric used to audit and backtest statistical demand forecasts.
ML-DSA-65	Module-Lattice Digital Signature Algorithm	NIST FIPS 204 (Dilithium3) post-quantum digital signature scheme combined in hybrid mode with Ed25519.
ML-KEM-768	Module-Lattice Key-Encapsulation Mechanism	NIST FIPS 203 (Kyber-768) post-quantum key-establishment scheme combined in hybrid mode with X25519.
MLS	Messaging Layer Security	IETF RFC 9420 protocol providing end-to-end encrypted group messaging inside councils.
MPC	Secure Multiparty Computation	Cryptographic paradigm compared against lightweight pairwise additive masking in Chapters 6–7.
MST	Sparse Merkle-Sum Tree	Authenticated tree binding both BLAKE3 hashes and non-negative supply/demand vector sums at every federation node.
NASA-TLX	NASA Task Load Index	Standardised human-factors metric used to measure cognitive load and participation fatigue in PoC testing.
NMT / TTS	Neural Machine Translation / Text-to-Speech	Local accessibility models translating qualitative Lists across languages and assisting visually impaired users.
OR	Operations Research	Discipline of mathematical optimisation, linear programming, and supply-chain clearing.
P2P	Peer-to-Peer	Decentralised networking architecture without central client-server gatekeepers.

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Abbreviation	Full Term	Definition & Role in Monograph
PoC	Proof of Concept	The 1,000-participant, 40-council, 80-good end-to-end reference implementation specified in Chapter 9.
PQ-UCAN	Post-Quantum User Controlled Authorisation Network	Chained, cryptographically signed capability certificates encoding council membership and revocable delegation.
PQC	Post-Quantum Cryptography	Cryptographic algorithms resistant to both classical and quantum (Shor/Grover) attacks.
PWA	Progressive Web Application	Offline-capable cross-platform web client running the Rust/WASM Role-LP engine on smartphones and desktops.
QUIC	Quick UDP Internet Connections	Modern encrypted, multiplexed transport protocol (quinn / iroh) used for Layer 2 federation sync.
Role-LP / CC / SR / PCW / BFT / GW	DPEP Node Roles 1–6	Lightweight Participant (LP), Council Coordinator (CC), Storage Replica (SR), Planning Compute Worker (PCW), Epoch Validator (BFT), and Delay-Tolerant Gateway (GW).
SLH-DSA	Stateless Hash-Based Digital Signature Algorithm	NIST FIPS 205 (SPHINCS+) signature standard relying solely on hash functions, used for long-term root anchors.
TPS	Transactions Per Second	Throughput metric shown in Chapter 5 to be irrelevant to epoch-synchronised participatory planning.
TTL	Time-To-Live	Hard expiration timer on delegate capability certificates and emergency replanning playbooks.
UniFFI	Universal Foreign Function Interface	Mozilla binding generator exposing the Rust dpep-core library to native Kotlin (Android) and Swift (iOS).
VRF	Verifiable Random Function	Cryptographic sortition primitive generating publicly verifiable random selections for audit and oversight boards.
WASM	WebAssembly	Deterministic, sandboxed binary instruction format (Wasmtime) executing IFB price-update rules.
ZKP	Zero-Knowledge Proof	Cryptographic proof verifying that a household's private consumption bundle obeys its budget $SC \leq B$ without revealing items.

C. Summary of Core Mathematical Notation (Chapter 4)

Symbol	Mathematical Meaning
Ω	Master catalogue of all $G = \lvert\Omega\rvert$ priced economic categories: $\mathcal{G}^{\text{priv}} \sqcup \mathcal{G}^{\text{pub}} \sqcup \mathcal{G}^{\text{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z}$.
$P^{(t)} \in \mathbb{R}_{+}^{\lvert\Omega\rvert}$	Vector of indicative social opportunity-cost prices at planning iteration t.
$\bar{a}_j, \underline{a}_j, r_j, e_j$	Worker council WC_j's proposed outputs ($\bar{a}_j$), intermediate inputs ($\underline{a}_j$), natural resources (r_j), and emissions (e_j).
$l_j^0, k_j^0, \Delta l_j, \Delta k_j$	Initial labour (l_j^0) and capital (k_j^0) endowments of WC_j, and proposed net additions/releases ($\Delta l_j, \Delta k_j$).

$r^*, l^*, k^*, E_z^{\max}$	Economy-wide available natural resources (r^*), labour supplies (l^*), capital stocks (k^*), and hard biophysical emission caps ($E_z^{\max}$).
$w_{\text{base}}, \epsilon_u, B_u(P)$	Base hourly remuneration rate (w_{base}), worker effort multiplier (ϵ_u), and citizen u's warranted annual consumption budget (B_u).
$S_i^{(t)}, D_i^{(t)}, SL_i^{(t)}$	Aggregate supply ($S_i^{(t)}$), aggregate demand ($D_i^{(t)}$), and planned slack buffer ($SL_i^{(t)} = \eta_i S_i^{(t)}$) for category i.
$\rho_i^{(t)}, \tau$	Normalized relative excess demand imbalance $\rho_i^{(t)} = \frac{2\lvert S_i^{(t)} - D_i^{(t)} \rvert}{S_i^{(t)} + D_i^{(t)}}$ and feasibility convergence threshold $\tau \in [0.01, 0.05]$.

CHAPTER 1

Executive Summary: Feasibility, Architecture, and Strategic Roadmap

> "No central agency composes the final plan. No market competition generates the final plan. A decentralised, social, iterative, communication process allows all actors to democratically formulate the final plan."
>
> — Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (1991), Ch. 6[1]

1.1 What Is Being Proposed

This monograph presents a rigorous, source-grounded feasibility study and technical architecture proposal for the Distributed Participatory Economics Protocol (DPEP): an open-source, local-first, cryptographically verifiable coordination protocol enabling democratic worker councils, neighbourhood consumer councils, and nested federations to formulate, iterate, ratify, and execute comprehensive economic plans without capitalist markets or authoritarian central planning.

The intellectual foundation of this protocol is Participatory Economics (Parecon), originally developed by Michael Albert and Robin Hahnel in The Political Economy of Participatory Economics (1991) and Looking Forward (1991),[2] elaborated across daily-life institutions and critic rebuttals in Michael Albert's Parecon: Life After Capitalism (2003),[3] and mathematically extended in Robin Hahnel's Democratic Economic Planning (2021),[4] Anders Sandström's Anarchist Accounting (2020),[5] and Mitchell Szczepanczyk's computer simulation work (pequod-cljs,

[1] Michael Albert and Robin Hahnel, Looking Forward: Participatory Economics for the Twenty First Century (Boston, MA: South End Press, 1991), Ch. 6 ("Participatory Allocation"), accessed via ZNetwork authorized online edition (https://znetwork.org/wp-content/uploads/zbooks/htdocs/ParEcon/lookingforward/lf6_1.htm).

[2] Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (Princeton, NJ: Princeton University Press, 1991), https://doi.org/10.2307/j.ctv131bvr2; Albert and Hahnel, Looking Forward (1991).

[3] Michael Albert, Parecon: Life After Capitalism (London & New York: Verso Books, 2003), full text archived at ZCommunications/ZNetwork (http://books.zcommunications.org/books/pareconv/).

[4] Robin Hahnel (with contributions by Peter Bohmer, Savvina Chowdhury, Allison Kerkhoff, Mitchell Szczepanczyk, and Michael Weisdorf), Democratic Economic Planning (Abingdon & New York: Routledge, 2021), https://doi.org/10.4324/9781003173700.

[5] Anders Sandström, Anarchist Accounting: Accounting Principles for a Democratic Economy (Abingdon & New York: Routledge, 2020), https://doi.org/10.4324/9781003036005; see also Sandström, "An Accounting System for a Libertarian Socialist Economy," Participatory Economy Project (Feb. 14, 2023), https://participatoryeconomy.org/an-accounting-system-for-a-libertarian-socialist-economy/.

2023).[6]

DPEP provides the computational substrate for four core institutional pillars:

1. Self-Managed Work and Consumption: Democratic worker councils (WC_j) and consumer councils (CC_h) in which participants exercise decision-making input proportional to the degree they are affected by an economic choice.[7]
2. Equitable Remuneration and Balanced Job Complexes (BJCs): Non-transferable consumption allowances calibrated to the duration, intensity, and onerousness (effort and sacrifice) of socially valued labour, supplemented by universal free public goods and council-approved special-need allowances, paired with job complexes balanced for empowerment and desirability within and across workplaces.[8]
3. Decentralised Iterative Planning: A multi-round coordination protocol in which councils submit production and consumption proposals, federated aggregators compute excess demands across seven economic categories (private goods, public goods, intermediate inputs, natural resources, labour categories, capital stocks, and pollutant emissions), and a deterministic Iteration Facilitation Board (IFB) engine updates indicative social opportunity-cost prices until a feasible, ecologically bounded plan is ratified.[9]
4. Intra-Year Operational Clearing and Scoped Replanning: Continuous tracking of deliveries and inventory drawdowns against the ratified plan, absorbing routine consumer preference shifts locally within neighbourhood and city slack buffers while mediating larger supply shocks across affected industry federations via Updating Facilitation Boards (UFBs).[10]

1.2 Why It Matters: Breaking the "No Third Way" Dogma

For over a century, mainstream and socialist economists alike have been haunted by Alec Nove's famous assertion in The Economics of Feasible Socialism (1983):

> "In a complex industrial economy the interrelation between its parts can be based in principle either on freely chosen negotiated contracts [i.e., markets],

[6] Mitchell Szczepanczyk, "Pseudocode and Algorithms for Computer Simulations of Democratically Planned Economies," Journal of Information Economics 1, no. 3 (2023): 43–54, https://doi.org/10.58567/jie01030004; open-source repository at https://github.com/msszczep/pequod-cljs.

[7] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 8, 16; Ch. 2, p. 24; Hahnel, Democratic Economic Planning, Ch. 1, pp. 14–16.

[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 25–30; Ch. 3, pp. 44–55.

[9] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, pp. 57–72; Ch. 5, pp. 73–106; Hahnel, Democratic Economic Planning, Ch. 7, pp. 130–162.

[10] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, pp. 66–67; Looking Forward, Ch. 6 (lf6_1.htm); Hahnel, Democratic Economic Planning, Ch. 8, pp. 167–170.

> or on a system of binding instructions from planning offices [i.e., command planning]. There is no third way."[11]

Yet historical experiments in democratic self-management—most notably the industrial and agrarian collectives of the Spanish Revolution (1936–1939), the Paris Commune (1871), federated cooperatives like Mondragón, and horizontal movements like Occupy Wall Street (2011)—repeatedly encountered two structural barriers when attempting to coordinate beyond single workplaces or rural villages:

- The Accounting and Valuation Barrier: Without a unified social accounting and indicative-price system, wealthier collectives hoarded superior inherited machinery (what Gaston Leval and Frank Mintz termed "collective egoism" or neocapitalismo colectivo in 1936 Barcelona),[12] while inter-industry coordination of complex intermediate inputs broke down into cumbersome bilateral barter.
- The Deliberative Scaling Barrier: Without structured, asynchronous computational protocols, horizontal deliberation degenerated into exhausting multi-hour plenary meetings with $\mathcal{O}(N^2)$ communication friction or unaccountable informal cliques (Jo Freeman's "tyranny of structurelessness").[13]

Contemporary distributed computing, post-quantum cryptography, and mathematical optimisation make it possible to overcome both barriers simultaneously—allowing millions of participants to coordinate complex industrial supply chains and ecological boundaries democratically in seconds of computation per round rather than months of bureaucratic paperwork.

1.3 Technical Feasibility: Achievable Guarantees vs. Aspirational Properties

A core mandate of our investigation is to distinguish demonstrably achievable engineering guarantees from aspirational or conditionally achievable properties requiring experimental validation.

[11] Alec Nove, The Economics of Feasible Socialism (London: George Allen & Unwin, 1983), pp. ix–x, 44; quoted directly at the opening of Chapter 4 in Albert and Hahnel, Political Economy of Participatory Economics, p. 57.

[12] Gaston Leval, Collectives in the Spanish Revolution, trans. Vernon Richards (London: Freedom Press, 1975); Frank Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, trans. Paul Sharkey (Oakland, CA: AK Press, 2013); Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), Ch. 6.

[13] Jo Freeman, "The Tyranny of Structurelessness," The Second Wave 2, no. 1 (1972) / Berkeley Journal of Sociology 17 (1972–1973): 151–164; David Graeber, The Democracy Project: A History, a Crisis, a Movement (New York: Spiegel & Grau, 2013).

[14] Leonid Hurwicz, "On Informationally Decentralised Systems," in Decision and Organisation, ed. C. B. McGuire and R. Radner (Amsterdam: North-Holland, 1972), pp. 297–336; Hurwicz, "The Design of Mechanisms for Resource Allocation," American Economic Review 63, no. 2 (1973): 1–30.

Table 1.1: Three-Tier Evidence Classification: Demonstrated Results, Analytical Estimates, and Research Targets

Category A: Empirically Demonstrated (Code / Data Audited)	Category B: Analytical Estimates (Schema / Complexity Models)	Category C: Open Research Targets (Require Phase 1–3 Validation)
1. Smooth Cobb-Douglas Convergence (pequod-cljs): Across 40 runs (dep1ex61..100.csv, 30k WCs/CCs, 500 goods), reaches $\rho_i \leq 5\%$ in 11.85 rounds (Year 1) and 6.53–6.58 rounds (Year 2), and $\rho_i \leq 3\%$ in 19.20 rounds.	1. Distributed Round Latency ($N = 10^6$): Aggregating 40k canonical CBOR proposals (208 MB raw / 60 MB compressed) across 50 servers + BFT checkpoint is estimated at 9–27 seconds/round (with 1.5–3× indexing/retry overhead).	1. Non-Convex & Leontief Convergence (Kill Gate 1): Whether Two-Tier Federation MILP + Convex Proximal-Jacobi ADMM clears Leontief + lumpy setup-cost economies in ≤ 25 rounds with $< 2\%$ MIP gap.
2. Refutation of Unregularised Tâtonnement on Scarf & Single-Plant Jumps: Fails Scarf's 3-good Leontief cycle ($\min \rho = 10.5\%$ over 3,000 rounds) and single indivisible plants ($\min \rho = 50.4\%$).	2. Shapley–Folkman Smoothing ($K \geq 20$): Disaggregating a 100-unit lumpy capacity across $K = 20$ heterogeneous councils shrinks the non-clearing gap from 50.4% to 0.40% (verify_economics.py).	2. Human Override & Agency (Kill Gate 1H): Whether participants in Phase 1H actively modify automated solver drafts ($\geq 30\%$ override rate) or passively rubber-stamp defaults.
3. NIST PQC & Prio3 VDAF Standards: FIPS 203/204/205 and IETF draft-irtf-cfrg-vdaf-16 (Prio3 secret-shared FLP proofs) are specified and standardised.	3. Mobile Round Bandwidth (Role-LP): 250–400 KB/round (145 KB for 43 validator sigs + 15–40 KB price deltas + 75 KB for $K = 3$ Prio3 shares + 512 B Merkle path).	3. Incentive & Soft-Budget Resistance: Mitigating capacity understatement (Weitzman 1980), soft-budget appeals (Kornai 1980), and carousel fraud (Hurwicz 1972).[14]

1.4 The Most Promising Architecture: Why We Reject a Global Blockchain

We explicitly reject building a per-transaction global blockchain (Architecture C), a centralised cloud database (Architecture A), or an unstructured gossip network without finality (Architecture B).

Participatory planning is not a stream of high-frequency financial trades; it is an epoch-synchronised, hierarchical bulk-synchronous coordination process followed by localized intra-year fulfilment. Furthermore, putting personal consumption requests on a public ledger violates personal privacy, and broadcasting 3.3 KB post-quantum signatures for every household action across all global nodes wastes bandwidth and energy.

Instead, our weighted evaluation (Chapter 6, scoring 4.51 / 5.00 vs. 2.54 / 5.00 for a blockchain) selects Architecture D — A Four-Layer Hybrid Local-First, Federated P2P, and Selective Epoch-BFT Architecture:

1. Layer 4 (Human Deliberation & Local Solver Edge): Accessible mobile/desktop Progressive Web App (PWA) and native clients running an embedded interior-point conic/quadratic solver (Clarabel.rs / HiGHS in WASM) for instant local proposal drafting and "what-if" collective-good exploration.

2. Layer 3 (Local-First Council & Household State): Encrypted local SQLite (SQLCipher) for structured vectors and accounting ledgers + Loro/Automerge CRDTs for offline-capable workplace and neighbourhood council deliberation, authorised via revocable post-quantum capability tokens (PQ-UCAN).
3. Layer 2 (Federated Aggregation & Content-Addressed Data Plane): Peer-to-peer QUIC (quinn / iroh) transport organising councils into nested Ward → City → Region and Industry Federation BLAKE3 Sparse Merkle-Sum Trees (MST), paired with content-addressed storage for qualitative council narratives (Lists).
4. Layer 1 (Selective Epoch-BFT Checkpoint & Deterministic IFB Engine): A lightweight rotating BFT validator committee (drawn from federated councils via Athenian-style cryptographic sortition) that commits only ∼10–20 global checkpoint blocks per annual planning cycle—anchoring each round's MST proposal root, executing the deterministic WASM price-update rule, and settling Sandström's macro Society Account.

1.5 Where AI Adds Value—and Where It Must Be Excluded

In strict adherence to Parecon's anti-coordinator principles,[15] AI must never become an unaccountable planning authority:

- 100% Deterministic Core: Indicative price adjustments, SB/SC ratios, and Society Account clearing use strictly deterministic fixed-point arithmetic and convex operations research (OR) solvers with zero neural-network weights.
- Accountable Edge Support: Classical statistical models (hierarchical ETS/ARIMA) forecast SKU size/color splits; local conic QP solvers (Clarabel.rs) automatically draft price-adjusted council revisions; and optional small local open-weight LLMs are restricted to multi-lingual translation, screen-reader accessibility, and summarising qualitative Lists with mandatory links to raw council source text.

1.6 Estimated Effort, Budget Scenarios, and the Five Key Decisions

The Five Most Important Decisions for Project Leadership

1. Commit to Architecture D (Local-First + Federated MST + Epoch Checkpoints) and Reject a Per-Transaction Blockchain.
2. Upgrade the Economic Solver from Unconstrained Cobb-Douglas to Constrained Conic/LP Optimisation with Proximal Damping (Algorithm 3).

[15] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 15–17; Looking Forward, Ch. 9 (lf9_2.htm) and Ch. 10 (lf10_1.htm).

Table 1.2: Summary of Implementation Roadmap and Budget Scenarios (2026 Verified Rates)

Scenario	Staff & Time	Infra / Audit	Total Est. Budget & Deliverable
Scenario A Solo + AI Coding	1.0 FTE 4–6 mos	\$150–\$300 \$0 (self-fuzz)	\$50k–\$90k (or \$0 cash + 1 FTE): Open-source PoC, mobile PWA, 100k benchmark.
Scenario B (Rec.) Small Research Team	3.5–4.5 FTEs 6–9 mos	\$2.5k–\$5k \$25k–\$45k	\$380k–\$700k: Production PoC, 1M benchmarks, 50-person pilot.
Scenario C Funded Co-op Pilot	10–14 FTEs 24–36 mos	\$35k–\$75k \$120k–\$200k	\$2.4M–\$4.3M: Verified v1.0 protocol, 10–30 co-op field pilot.

3. Enforce Hard Biophysical Caps ($r^*, E_z^{\max}$) Lexicographically Above PDRM Pollution Pricing.
4. Design for a "3-Milestone Hybrid Deliberation Schedule" separating automated intermediate input-clearing sub-iterations from human value-deliberation milestones.
5. Target Dual Utility: Support both full-scale Parecon simulation research and immediate deployment for federated worker cooperatives and participatory budgeting communities ("Experiment 2: Developing a Parallel Economy").[16]

Final Recommendation: PROCEED WITH PHASE 1 AND PHASE 1H AS STRICT KILL GATES—implementing the Python-first Economic Coordination & Non-Convexity Simulator (Phase 1) and the Low-Technology Human Participation Probe (Phase 1H) first, and holding all Phase 2 (distributed Rust nodes) and Phase 3 (post-quantum & Prio3 cryptographic field pilots, requiring a mandatory \$50k–\$150k external security audit) strictly contingent on passing Kill Gates 1 and 1H.

[16] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 129–130.

CHAPTER 2

Historical and Institutional Inspirations: From Council Practice to Protocol Design

> "In almost all the industries, factories, mills, workshops, transportation services, public services, and utilities, the rank and file workers, their revolutionary committees, and their syndicates reorganized and administered production, distribution, and public services without capitalists, high salaried managers, or the authority of the state… They instituted not bourgeois formal democracy but genuine grass roots functional libertarian democracy, where each individual participated directly in the revolutionary reorganization of social life."
>
> — Gaston Leval, Espagne Libertaire (1971), p. 11, translated in Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (1974), Ch. 1, p. 6[1]

2.1 Epistemological Premise: Institutions Before Cryptography

A recurring failure mode of contemporary "decentralised governance" (DAO) architectures is the conflation of cryptographic trustlessness with democratic self-management. Recording a tally on an immutable ledger does not render an economy democratic if:

1. Proposal drafting and agenda-setting are monopolised by an unaccountable technical or managerial elite—what Albert and Hahnel diagnose as the coordinator class.[2]
2. Decision-making influence is proportional to token wealth or meeting endurance rather than proportional to the degree participants are affected.
3. Wealthier production collectives hoard superior inherited productive endowments and refuse to equalise resources across regions.

In this chapter, we examine five historical and contemporary institutional traditions as empirical laboratories of democratic coordination—analyzing both their institutional breakthroughs and their structural failures—to derive binding architectural rules for DPEP.

[1] Gaston Leval, Espagne Libertaire (1936–1939) (Paris: Éditions du Monde Libertaire, 1971), p. 11; verbatim English translation in Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), Ch. 1, p. 6.

[2] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 14–17, 20–22; Looking Forward, Prologue ("Coordinatorism" and "The Origins of Coordinatorism").

2.2 The Spanish Revolution (1936–1939): Industrial and Agrarian Collectives

Following the defeat of the Nationalist military coup in July 1936 across Catalonia, Aragon, Valencia, and parts of Castile and Andalusia, workers and peasants affiliated with the anarchist CNT (Confederación Nacional del Trabajo) and socialist UGT (Unión General de Trabajadores) collectivized roughly 70–80% of industry in Catalonia and over 400 agrarian collectives in Aragon. Quantitative demographic studies by Frank Mintz place participation at a minimum of $1,838,000$ collectivists ($1,080,000$ industrial and $758,000$ agrarian, revised in 1977 from his earlier 1970 estimate of 2.44–3.2 million), while eyewitness researcher Gaston Leval estimated up to 3 million participants.[3]

2.2.1 Internal Self-Management, Delegation, and Recall

Across both Barcelona factories (tramways, textiles, metalworks, water/gas/electric utilities) and Aragonese villages (Alcañiz, Calanda, Binéfar, Graus, Mas de las Matas), supreme authority resided in the weekly or biweekly General Assembly (Asamblea General) of all workers or villagers.[4] Assemblies elected specialized administrative committees (Comités de Administración) to manage daily coordination, input procurement, and statistical reporting. As documented by Bookchin, Peirats, and Leval, these delegates operated under strict anti-bureaucratic constraints:

> "Where such popular assemblies were possible, their decisions were executed by a committee elected from the assembly. Apparently, the right to recall committee members was taken for granted and they certainly enjoyed no privileges, emoluments, or institutional power… It remained a cardinal principle of Spanish anarchists never to pay their delegates, even when the CNT numbered a million members. Normally, the responsibilities of elected delegates had to be discharged after working hours."[5]

2.2.2 Inter-Collective Coordination, Reorganization, and Accounting

- Regional Equalization in Aragon: Established in October 1936 (and formalised at the Congress of Caspe in February 1937 representing ∼450 collectives and $300,000$ collectivists), the Federation of Agrarian Collectives of Aragon pooled

[3] Frank Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, trans. Paul Sharkey (Oakland, CA: AK Press, 2013 [1970/1977]); Gaston Leval, Collectives in the Spanish Revolution (London: Freedom Press, 1975); Sam Dolgoff, ed., The Anarchist Collectives: Workers' Self-Management in the Spanish Revolution, 1936–1939 (New York: Free Life Editions, 1974), available in full at https://theanarchistlibrary.org/library/sam-dolgoff-editor-the-anarchist-collectives.

[4] Dolgoff, The Anarchist Collectives, Ch. 2 and Ch. 9; Murray Bookchin, "Introductory Essay," in Dolgoff, The Anarchist Collectives, pp. xxvii–xxx.

[5] Bookchin, "Introductory Essay," in Dolgoff, The Anarchist Collectives, p. xxviii.

surplus wheat, olive oil, and livestock at cantonal depots, transferred machinery and seed from fertile to arid villages, and conducted direct inter-regional trade with Catalonian textile syndicates without commercial middlemen.[6]

- Unified Export Coordination in Valencia (CLUEA): In the Levante, the CNT and UGT established the Consejo Levantino Unificado de Exportación Agrícola (CLUEA), federating ∼270 local orange and citrus committees to standardize grading, rail/maritime transport, and export clearing.[7]
- Industrial Recomposition in Catalonia: In Barcelona, the CNT's industrial unions (Sindicatos Únicos de Ramo) consolidated hundreds of small, unsanitary workshops in woodworking, baking, tanning, and optic/glass production into modern integrated plants, eliminating redundant middlemen while preserving full employment.[8]
- Accounting and the "Family Wage": Many Aragonese villages abolished official pesetas internally, distributing abundant staples free according to need (Maxim 4) and issuing a family consumption booklet (libreta de consumo) or local points calibrated to family size for scarce goods.[9] Inter-collective trade relied on standardised valuation tables (indexed in wheat equivalents or shadow pesetas) cleared through cantonal ledgers.

2.2.3 Historical Limitations and Structural Failures

Despite remarkable wartime productivity in munitions, transport, and agriculture, the Spanish collectives suffered three critical weaknesses that directly inform DPEP's design:

1. "Collective Egoism" (Neocapitalismo Colectivo) Between Rich and Poor Enterprises: In urban Catalonia, prior to the socialisation decrees of late 1937, collectives remained financially autonomous inside a quasi-market environment. Wealthy collectives that inherited modern machinery or lucrative monopolies (such as the Barcelona tramways, power utilities, and luxury hotels) enjoyed large surpluses, while raw-material-starved textile and construction collectives struggled to pay base wages.[10] Protocol Lesson: This historical failure proves why Albert and Hahnel (1991a, Ch. 5, pp. 84–87) and Sandström (2020) insist that productive assets belong to the Productive Commons, and that every Worker Council must be charged an opportunity-cost rental fee ($P_k \cdot k_j$) on its inherited capital stock!

[6] Leval, Collectives in the Spanish Revolution, Ch. 6; Dolgoff, The Anarchist Collectives, Ch. 10.
[7] Mintz, Anarchism and Workers' Self-Management in Revolutionary Spain, Ch. 4.
[8] Augustin Souchy, "Collectivizations in Catalonia," in Dolgoff, The Anarchist Collectives, Ch. 7.
[9] Leval, Collectives in the Spanish Revolution, pp. 165–210.
[10] José Peirats, "Pay Scales and Collectible Egoism," in Dolgoff, The Anarchist Collectives, Ch. 6; Mintz, Anarchism and Workers' Self-Management.

2. Burnout from Unpaid Evening Governance: Requiring delegates to perform all federation administration after a full day of physical factory labour caused severe exhaustion and favoured childless militants over workers with caregiving duties. Protocol Lesson: Governance and facilitation work must be counted inside paid work hours as part of every citizen's Balanced Job Complex.
3. Lack of Unified Shadow Pricing for Complex Intermediate Chains: While cantonal barter cleared simple agricultural staples, urban industries lacked an iterative mechanism to compute social opportunity costs across hundreds of interdependent intermediate inputs once pre-war market prices grew stale.
4. Wartime Militia Coercion, Smallholder Conflict, and "Workers Against Work" (Casanova 2005; Seidman 1991): Modern archival historiography demonstrates that internal weaknesses were not merely accounting bugs. As Julián Casanova documents, several rural Aragonese collectives formed under the coercive shadow of armed militia columns, provoking acute conflict with "individualist" peasant smallholders.[11] Inside Barcelona's collectivised factories, Michael Seidman documents persistent worker absenteeism, wage demands, and resistance to industrial work discipline under wartime inflation—while externally, the collectives were starved of credit by the central government, physically dissolved in Aragon by Enrique Líster's 11th Division in August 1937, and crushed by Franco's victory in 1939.

2.3 The Paris Commune (March 18 – May 28, 1871)

During its 72 days of existence before its bloody suppression during La Semaine Sanglante, the Paris Commune established foundational principles of council democracy celebrated in Karl Marx's The Civil War in France (1871).[12] A rigorous institutional analysis must distinguish what the Commune actually implemented from unimplemented proposals and later theoretical extrapolations.

2.4 Classical Athenian Democracy (∼508/7 – 322 BCE)

Over nearly two centuries from the reforms of Kleisthenes (508/7 BCE) to the Macedonian conquest (322 BCE), classical Athens developed the most sophisticated institutional machinery for direct citizen self-governance in antiquity, ana-

[11] Julián Casanova, Anarchism, the Republic and Civil War in Spain: 1931–1939 (London & New York: Routledge, 2005), Chs. 6–8; Michael Seidman, Workers Against Work: Labour in Paris and Barcelona during the Popular Fronts (Berkeley: University of California Press, 1991), Chs. 3–6; Seidman, Republic of Egos (2002).

[12] Karl Marx, The Civil War in France (London: General Council of the International Workingmen's Association, 1871); John Merriman, Massacre: The Life and Death of the Paris Commune of 1871 (New York: Basic Books, 2014).

[13] Marx, The Civil War in France, Section III.

[14] Merriman, Massacre: The Life and Death of the Paris Commune of 1871, Ch. 5.

Table 2.1: Paris Commune (1871): Actually Implemented Institutions vs. Proposals & Extrapolations

Dimension	Actually Implemented (March–May 1871)	Proposed on Paper or Later Extrapolation
Representation & Recall	Municipal elections across 20 arrondissements (March 26, 1871) by universal male suffrage; councilors declared revocable at any time; merged legislative and executive work across 10 Commissions.[13]	Formal recall was rarely exercised during the 72-day military siege; women were excluded from formal voting despite leading grass-roots defence and cooperative organising via the Union des Femmes (Élisabeth Dmitrieff, Nathalie Lemel).
Remuneration Cap	Decree of April 2, 1871: Capped maximum salaries of all Commune officials and municipal administrators at 6,000 francs/year (equivalent to a skilled Parisian workman's wage) and banned multiple stipends.	Did not abolish wage hierarchies inside private Parisian enterprises or establish balanced job complexes inside workshops.
Workplace Self-Management	Decree of April 16, 1871 (drafted by Léo Frankel's Commission of Labour and Exchange): Ordered a census of workshops abandoned by fleeing owners and their transfer to worker cooperative associations; abolished night baking (April 20) and employer wage fines (April 27).[14]	Only a few dozen cooperative workshops (e.g., the Louvre armaments workshop electing its own council and foremen; women's uniform cooperatives) actually began operating before Versailles troops entered Paris on May 21.
National Federation & Credit	Operated strictly within besieged Paris; left the Banque de France intact (negotiating advances via Charles Beslay rather than socializing credit).	Declaration to the French People (April 19, 1871) (drafted by Proudhonian Pierre Denis) and Marx envisioned a national federation of autonomous communes coordinating via delegated national assemblies—never realized due to the crushing of provincial communes in Lyon and Marseille.
Crisis Centralisation	On May 1, 1871, under bombardment, the Jacobin/Blanquist majority voted (45 to 23) to create a 5-member Committee of Public Safety with sweeping powers over the objection of the Federalist minority (Frankel, Varlin, Longuet, Courbet).	Demonstrates the acute vulnerability of council democracies during crisis to revert to emergency command authority unless crisis-mode bounds (TTLs and post-incident audits) are constitutionally hard-coded.

lyzed definitively by Mogens Herman Hansen.[15]

2.4.1 Core Institutional Mechanisms Adaptable to DPEP

1. Direct Assembly (Ekklesia) vs. Norm-Setting (Nomothetai): Open to all adult male citizens (∼30,000–40,000 eligible in the 4th century BCE, with ∼6,000 attending the 40 annual meetings on the Pnyx), the Ekklesia voted on decrees (psephismata), while permanent constitutional laws (nomoi) were ratified by a separate panel of 1,000 nomothetai selected by lot.
2. Sortition (Kleroterion) and Strict Rotation: Roughly 600 of the ∼700 domestic magistrates (archai) and the 6,000 annual jurors (dikastai) were selected by lot using bronze allotment machines (kleroteria) rather than election—following Aristotle's principle (Politics 1294b) that elections are intrinsically oligarchic (favoring the wealthy, famous, and rhetorically polished) whereas sortition is democratic.[16] Almost all offices were restricted to a non-repeatable one-year term (or repeatable only once in a lifetime for the Boule), preventing a permanent coordinator class.
3. Probouleutic Agenda Facilitation (Boule of 500): Five hundred citizens (50 from each of the 10 Kleisthenic tribes, chosen by lot from local demes) served for one year on the Council (Boule), with each tribe serving as the standing executive committee (prytaneis) for 1/10th of the year and a new chair (epistates) drawn by lot every 24 hours. Crucially, the Boule's role was probouleutic: it prepared structured proposals (probouleumata) for the Ekklesia to debate, amend, or reject, but could not impose major policies on its own.[17] This is the direct institutional ancestor of Parecon's Facilitation Boards.
4. Public Scrutiny and Mandatory Audit (Dokimasia, Euthynai, Graphe Paranomon): Every magistrate underwent pre-office eligibility scrutiny (dokimasia) and a mandatory end-of-term financial and conduct audit (euthynai) before 10 logistai (accountants) and 10 euthynoi chosen by lot.[18]
5. Civic Remuneration (Misthophoria): Introduced by Pericles for jurors and councilors and expanded in the 4th century BCE to assembly attendance (misthos ekklesiastikos, 3 to 6 obols per day), ensuring working citizens (thetes) could afford to participate.

2.4.2 Explicit Examination of Structural Exclusions

Athenian democracy excluded women (who lacked political citizenship and independent property control), metics (resident non-citizens, ∼20,000–30,000 artisans

[15] Mogens Herman Hansen, The Athenian Democracy in the Age of Demosthenes: Structure, Principles, and Ideology (Oxford: Blackwell / Norman: University of Oklahoma Press, 1991).
[16] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 9, pp. 230–245.
[17] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 10, pp. 246–265.
[18] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 9, pp. 218–224.

and merchants), and enslaved people (∼80,000–150,000 forced laborers in the Laurion silver mines, agriculture, crafts, and households), leaving only ∼10–15% of the total population with full political rights.[19] Furthermore, as Moses Finley and G. E. M. de Ste. Croix demonstrate, Athens' fiscal capacity to fund misthophoria and public works depended heavily on the Laurion silver mines (worked by tens of thousands of enslaved miners) and 5th-century Delian League imperial tribute (phoros).[20] In a universal, non-extractive participatory economy, every hour spent in deliberation and euthynai audits carries a real opportunity cost in productive labour; governance time must therefore be budgeted inside paid Balanced Job Complexes and economised carefully, while sortition uses commit-reveal hash beacons across all citizens regardless of hardware ownership.

2.5 Occupy Wall Street (2011) and Additional Precedents

- Occupy Wall Street (2011), Freeman's "Structurelessness," and Assembly Scaling: In small groups without formal rules, Jo Freeman (1972) showed that power concentrates in unaccountable informal friendship cliques ("the tyranny of structurelessness")—a sociological dynamic that asynchronous software alone does not eliminate.[21] Independently, Mansbridge (1980) and Dahl & Tufte (1973) show that synchronous assemblies above ∼50–150 participants face steep $\mathcal{O}(N)$ serial speaking-time and $\mathcal{O}(N^2)$ interpersonal deliberation limits, while Albert (2003, Ch. 5) shows why unanimity veto rules fit small-team hiring but majority/supermajority rules with minority reconsideration delays fit workplace investments.
- Mondragón Cooperative Corporation (1956–Present): Across ∼90 cooperatives and ∼70,000–80,000 total workers (of whom ∼30,000–35,000 are cooperative worker-owners, socios, while 60–70% of retail Eroski and international subsidiary staff are non-member wage employees), Mondragón separates the elected financial Audit / Watchdog Committee (Comisión de Vigilancia) from the frontline workers' Social Council (Consejo Social),[22] illustrating both federated industrial self-management and the erosion of cooperative membership under global market competition.
- Porto Alegre Participatory Budgeting (1989–2017 Suspension): Combined regional assemblies and recallable delegates with an objective Quality of Life

[19] Hansen, Athenian Democracy in the Age of Demosthenes, Ch. 4, pp. 86–124.

[20] M. I. Finley, Economy and Society in Ancient Greece (London: Chatto & Windus, 1981), Ch. 5; G. E. M. de Ste. Croix, The Class Struggle in the Ancient Greek World (Ithaca, NY: Cornell University Press, 1981).

[21] Graeber, The Democracy Project, Chs. 1–3; Freeman, "The Tyranny of Structurelessness," pp. 151–164; Jane J. Mansbridge, Beyond Adversary Democracy (New York: Basic Books, 1980); Robert A. Dahl and Edward R. Tufte, Size and Democracy (Stanford: Stanford University Press, 1973); Albert, Parecon: Life After Capitalism (2003), Ch. 5.

[22] William Foote Whyte and Kathleen King Whyte, Making Mondragón, 2nd ed. (Ithaca, NY: ILR Press, 1991); Fred Freundlich, The Mondragon Cooperative Experience (2015).

Need Matrix to redirect municipal investment to poor districts, before being suspended in 2017 following municipal coalition shifts, fiscal austerity, and ∼2,400 unexecuted citizen project requests.

- Rojava / DAANES (2012–2026 Damascus Integration Accord): Implemented nested communes, co-chair gender parity, and autonomous women's councils (Kongra Star),[23] before military and geopolitical pressure led to the 30 January 2026 SDF–Damascus integration agreement merging DAANES civil and security institutions into the Syrian state.
- Ostrom's Common-Pool Resource (CPR) Institutions (1990): Elinor Ostrom's study of enduring self-governed commons (e.g., Valencia's Tribunal de las Aguas) established eight design principles: defined boundaries, proportional equivalence, collective-choice arrangements, accountable monitoring, graduated sanctions, rapid local conflict resolution, right to organise, and nested enterprises.[24]

[23] Hahnel, Bohmer, and Chowdhury, "Reproductive Labour," Ch. 10 in Hahnel, Democratic Economic Planning, p. 200.

[24] Elinor Ostrom, Governing the Commons (Cambridge: Cambridge University Press, 1990), Ch. 3, p. 90.

Synthesis: Comparative Institutional-to-Protocol Matrix

Table 2.2 synthesizes the eight core institutional mechanisms analyzed across Chapters 2.2–2.5, mapping their historical strengths and limitations directly to concrete capabilities in the Distributed Participatory Economics Protocol (DPEP).

Table 2.2: Comparative Matrix: Institutional Mechanisms Mapped to DPEP Protocol Capabilities

Mechanism & Precedents	Strengths & Advantages	Historical Limitations	Concrete DPEP Protocol Capability
1. General Assembly Spain 1936; Athens Ekklesia; Occupy; Mondragón	Builds face-to-face solidarity; surfaces qualitative context.	Fails above ∼50–150 speakers ($\mathcal{O}(N^2)$ friction); meeting fatigue.	Scoped strictly to work teams (10–30) and neighborhood councils (50–200) with async pre-reading queues.
2. Revocable Delegation Paris Commune 1871; Spain CNT; Porto Alegre PB	Enables multi-tier federation scaling while tethering delegates to base.	Delegates ossify if recall is procedurally slow or difficult.	Scoped PQ-UCAN Delegate Certificates: Issued by council threshold sig; revocable in a single council vote.
3. Sortition + Rotation Classical Athens (Boule, Logistai, Dikasteria)	Prevents factional elections, charismatic capture, and coordinator elites.	Unsuitable for rare specialist tasks unless drawn from qualified pool.	VRF Cryptographic Sortition: Selects rotating IFB oversight, audit panels, and dispute juries.
4. Mandatory Audit Athens Euthynai (Logistai); Mondragón Consejo Social	Systematically detects bias or unauthorized overrides.	Requires tamper-evident records and independent auditors.	Deterministic WASM + Merkle-Sum Audit Logs: Any node can verify IFB price updates in < 10 ms.
5. Endowment Equalization Aragon Depots (1936); Mondragón Funds	Prevents collectives with superior machinery from extracting rents.	Failed in 1936 Barcelona where rich collectives hoarded revenues.	Stand-Alone Capital Asset Ledger: Automatically debits each WC_j's SC_j for capital/resource user rights (P_k, P_r).
6. Proportional & Caucus Rojava Co-Chairs & Women's Councils	Protects disproportionately affected groups from majority tyranny.	Can deadlock if scope of veto is unbounded.	Impact-Weighted & Dual-Key Governance Rules (PEPE Ch. 2 p. 24; DEP Ch. 10 p. 200).
7. Iterative Tâtonnement IETF/Git; Lange-Arrow-Hurwicz; Parecon IFB	Scales quantitative clearing to millions asynchronously.	Pure prices strip out human context unless paired with qualitative text.	Federated Sparse Merkle-Sum Trees + Content-Addressed Qualitative Lists.
8. Graduated Sanctions Ostrom CPRs (Valencia Tribunal de las Aguas)	Resolves 95% of operational friction locally with minimal overhead.	Requires clear invariants and fast local mediation.	Tiered Variance Alerts → UFB Mediation → Industry/Environmental Court.

CHAPTER 3

Source-Grounded Requirements of Participatory Economics

> "Until now, however, descriptions of alternatives have been unconvincing. Here Michael Albert and Robin Hahnel support the libertarian socialist tradition by presenting a rigorous, well-defined model of how producers and consumers could democratically plan their interconnected activities."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Front Flap / JSTOR Abstract[1]

3.1 Source Provenance and Classification Methodology

Before formulating mathematical equations or distributed protocols, we establish strict traceability to the foundational texts. Every requirement in Table 3.1 is classified into one of four provenance tiers:

- [ORIGINAL-1991]: Explicitly specified in Michael Albert and Robin Hahnel's two 1991 volumes, The Political Economy of Participatory Economics (PEPE) and Looking Forward (LF).[2]
- [INTERPRETATION-LATER]: Institutional elaborations, formalizations, and accounting procedures introduced in Michael Albert's Parecon: Life After Capitalism (PLAC, 2003), Robin Hahnel's Democratic Economic Planning (DEP, 2021) and A Participatory Economy (2022), Anders Sandström's Anarchist Accounting (2020/2023), Mitchell Szczepanczyk's simulation work (2023), and the Participatory Economy Project's User Story Mapping (2023).[3]
- [TECH-EXTENSION]: Technological extensions required to execute the model over a modern, decentralised, post-quantum, privacy-preserving computational network.
- [UNRESOLVED]: Open theoretical or institutional questions requiring simulation or empirical human research (examined in full in Chapter 10).

[1] Albert and Hahnel, Political Economy of Participatory Economics, book overview (https://www.jstor.org/stable/j.ctv131bvr2).

[2] Albert and Hahnel, Political Economy of Participatory Economics (1991), print pagination verified via library TOC scan (research/sources/pepe_dandelon.pdf, pp. 1–132) and full text analyzed via ZNetwork (intro.htm, 1.htm–6.htm); Albert and Hahnel, Looking Forward (1991), full text analyzed via ZNetwork (lfIntroduction.htm, lf1_1.htm–lf10_1.htm).

[3] Hahnel, Democratic Economic Planning (2021); Robin Hahnel, A Participatory Economy (Chico, CA: AK Press, 2022); Sandström, Anarchist Accounting (2020); Szczepanczyk, "Pseudocode and Algorithms" (2023); Participatory Economy Project, "Participatory Planning: User Story Mapping" (2023), https://participatoryeconomy.org/project/participatory-planning-user-story-mapping/.

3.2 Normative Foundations and Endogenous Preference Theorems

In Chapter 1 of The Political Economy of Participatory Economics (pp. 8–11), Albert and Hahnel evaluate economic institutions against five valuative criteria: Equity, Self-Management, Solidarity, Variety, and Efficiency.[4] For distributive equity, they distinguish four mutually exclusive maxims (PEPE Ch. 1, p. 8; Ch. 3, p. 44; DEP Ch. 1, pp. 16–22):

1. Maxim 1 (Capitalism): Payment according to the value of the contribution of one's personal labour and productive property owned (rejected due to inheritance—the "Rockefeller grandson problem"—luck, and compounding wealth accumulation).[5]
2. Maxim 2 (Market Socialism / Coordinatorism): Payment according to the value of one's personal labour contribution alone (rejected because differences in marginal productivity stem overwhelmingly from the "genetic lottery," luck, job assignment, and publicly funded training rather than personal sacrifice—illustrated by their "Mozart vs. Salieri" and "Dave Butz" examples).[6]
3. Maxim 3 (Participatory Economics): Payment according to effort and personal sacrifice expended in socially valued labour (longer hours, higher intensity, or more onerous/dangerous conditions prior to full BJC equalization).
4. Maxim 4 (Humane Solidarity): Payment according to need—incorporated into Parecon via universal free public goods (healthcare, education) and council-approved special-need allowances (PEPE Ch. 3, p. 47).

Crucially, building on their mathematical proofs in Quiet Revolution in Welfare Economics (1990),[7] Albert and Hahnel prove that because human preferences are endogenous (people rationally mold their future characteristics ΔC in anticipation of future terms of supply), any institution that systematically mischarges for activities relative to true social opportunity cost causes: (1) a snowballing divergence from Pareto optimality over time, (2) systematic warping of human development trajectories, and (3) ex post invisibility of the bias to participants. Markets systematically underprice negative externalities and overprice public goods (warping preferences toward anti-social individualism), while central planning systematically under-supplies self-managed work roles (warping workers toward apathy and coordinator submission).[8]

[4] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 8–9, 11.
[5] Hahnel, Democratic Economic Planning, Ch. 1, pp. 16–18; Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 45–46.
[6] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 46–47.
[7] Robin Hahnel and Michael Albert, Quiet Revolution in Welfare Economics (Princeton, NJ: Princeton University Press, 1990), Chs. 4, 6, 7; summarised in Political Economy of Participatory Economics, Ch. 1, pp. 9–11.
[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 1, pp. 12–17; Ch. 5, pp. 98–105.

3.3 Formal Traceability Matrix of Requirements

Table 3.1: Master Requirements Matrix Derived from Primary and Secondary Literature

Req ID	Requirement Specification	Classification	Primary Citations
REQ-NORM-01	Productive Commons & User-Right Fees: Abolish private ownership of productive property; all natural resources and capital stocks belong to society. Worker councils acquire temporary user rights during annual planning and pay opportunity-cost charges (P_r, P_k, P_l).	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 1 (pp. 17–20), Ch. 5 (p. 85); DEP Ch. 6 (pp. 98–102), Ch. 12 (p. 244); Sandström (2020).
REQ-NORM-02	Proportional Self-Management: Decision-making input proportional to the degree affected by an economic choice.	[ORIGINAL-1991]	PEPE Ch. 1 (p. 8), Ch. 2 (p. 24); LF Ch. 1; PLAC Ch. 5; DEP Ch. 1 (p. 14).
REQ-NORM-03	Effort & Sacrifice Remuneration: Consumption budgets proportional to effort/sacrifice (Maxim 3) tempered by universal free goods and need exceptions (Maxim 4), with council average effort ratings capped by SB_j/SC_j.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 3 (pp. 44–55); DEP Ch. 6 (pp. 108–117); Sandström (2023).
REQ-NORM-04	Balanced Job Complexes (BJCs): Tasks partitioned into subsets $\{PIPA\}_i$ ordered by empowerment/desirability and combined into balanced complexes within and across workplaces via Job Complex Committees.	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 25–30), Ch. 5 (pp. 82–83); LF Chs. 1–2.
REQ-ACT-01	Nested Consumer Councils & Federations: Individuals/households → Neighbourhood CC_h → Ward → City/County → State/Region → National Federation, requesting public goods matched to spatial scale.	[ORIGINAL-1991]	PEPE Ch. 3 (pp. 40–42), Ch. 5 (pp. 81–82); LF Chs. 3, 4, 8.
REQ-ACT-02	Worker Councils & Dual Federations: Democratic workplace councils federated both by industry (scale economies, R&D, SB/SC benchmarking) and by region (local production externalities).	[ORIGINAL-1991]	PEPE Ch. 2 (pp. 23–25), Ch. 4 (p. 62), Ch. 5 (p. 88); LF Chs. 1, 2, 7.
REQ-ACT-03	Six Functional Facilitation Boards: Iteration (IFB), Consumption (CFB), Production (PFB), Employment (EFB), Household (HFB), and Updating (UFB) facilitation boards performing purely informational/mediating tasks.	[ORIGINAL-1991]	LF Ch. 6 (lf6_1.htm), Ch. 9 (lf9_2.htm); PEPE Ch. 4 (p. 62).
REQ-ACT-04	Communities of Affected Parties (CAPs) & Environmental Courts: Airshed/watershed collectives participating in the Pollution Demand-Revealing Mechanism (PDRM) below hard biophysical caps.	[INTERP-LATER]	DEP Ch. 7 (pp. 138–149), Ch. 14 (pp. 261–267); PEP (2023).

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Req ID	Requirement Specification	Classification	Primary Citations
REQ-ACT-05	Stand-Alone Capital Asset Entities & Society Account: Double-entry social accounting entities tracking capital production costs, user-right fees, CAP flows, and GDP division between consumption and investment.	[INTERP-LATER]	Sandström (2020, Chs. 3–5; 2023).
REQ-DATA-01	Seven Priced Economic Categories: Private goods, public goods, intermediate inputs, natural resources, labour categories, capital stocks, and regulated pollutants.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 5 (pp. 78–88); DEP Ch. 7, Ch. 9; Szczepanczyk (2023).
REQ-DATA-02	Coarse Planning Taxonomy vs. Granular SKUs: Plan over coarse resource-equivalent product classes; extrapolate specific sizes/colors statistically and resolve at intra-year fulfilment.	[ORIGINAL-1991] + [INTERP-LATER]	PEPE Ch. 4 (p. 64); LF Ch. 6 (lf6_2.htm); DEP Ch. 8 (pp. 163–167).
REQ-DATA-03	Public Books of Blueprints & Qualitative Lists: Publish all technical input-output recipes $[A, K, R, L]$ and attach qualitative descriptive Lists to quantitative council vectors.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 61–62), Ch. 5 (p. 84), Ch. 6; PLAC Chs. 11, 17.
REQ-DATA-04	Protected Private Consumption Sphere: Personal consumption requests within warranted budget B_u cannot be vetoed by neighbours and must be cryptographically shielded from neighbour snooping.	[ORIGINAL-1991] + [TECH-EXT]	PEPE Ch. 3 (p. 41), Ch. 5 (p. 80); LF Ch. 3; PLAC Chs. 12, 18; Dapprich (2024).
REQ-PLAN-01	Multi-Horizon & Collective-First Sequencing: Long-run (education, environment) and aggregate investment plans precede annual planning; federation collective goods are settled before final private consumption bundles.	[ORIGINAL-1991] + [INTERP-LATER]	LF Ch. 3 (lf3_2.htm), Ch. 8 (lf8_1.htm), Ch. 9 (lf9_1.htm); DEP Chs. 11–15.
REQ-PLAN-02	Slack Planning & Hierarchical Intra-Year Clearing: Include 3–5% planned excess supply buffer (SL^E); cancel routine consumer swaps inside neighbourhood/ward/city federations; trigger UFB replanning only when net shocks exceed slack.	[ORIGINAL-1991]	PEPE Ch. 4 (pp. 66–67), Ch. 6 (p. 113, 122); DEP Ch. 8 (pp. 167–170).

CHAPTER 4

Formal Economic Model, Accounting Identities, and Algorithms

> "It is impossible to show that a feasible nonmarket system at least approaches the productivity of the market unless (1) a rather well-developed theoretical model of the nonmarket system is available, and (2) it is demonstrated that a sufficiently productive approximation of the ideal… system described in the theoretical model is practically possible."
>
> — Allen Buchanan, Ethics, Efficiency, and the Market (1985), p. 29, quoted in Albert and Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 58, and Ch. 5, p. 73[1]

4.1 Ontology of Economic Actors and Priced Objects

Let the participatory economy $\mathcal{E}$ over annual planning period T consist of the following finite sets of actors and institutional entities:[2]

- Individuals ($\mathcal{I} = \{1, \ldots, N_{\text{ind}}\}$): Natural persons participating as consumers, workers, and citizens. Each individual $u \in \mathcal{I}$ belongs to a primary neighbourhood consumer council $h(u) \in \mathcal{H}$ and, if working, a primary worker council $j(u) \in \mathcal{J}$ (plus optional secondary council $j'(u)$ for cross-workplace Balanced Job Complex rotation).[3]
- Neighbourhood Consumer Councils ($\mathcal{H} = \{CC_1, \ldots, CC_H\}$) & Nested Consumer Federations ($\mathcal{F}^{\text{C}} = \bigcup_{\ell=1}^{L} \mathcal{F}_\ell^{\text{C}}$): A rooted geographic hierarchy of depth $L = 5$ (Level 1 = Ward, Level 2 = City/County, Level 3 = State, Level 4 = Region, Level 5 = National Federation).[4]
- Worker Councils ($\mathcal{J} = \{WC_1, \ldots, WC_J\}$) & Dual Worker Federations ($\mathcal{F}^{\text{W}} = \mathcal{F}^{\text{W,ind}} \cup \mathcal{F}^{\text{W,reg}}$): Democratic workplace collectives federated both by industry ($IF_k \in \mathcal{F}^{\text{W,ind}}$) and by geographic region ($\mathcal{F}^{\text{W,reg}}$).[5]

[1] Allen Buchanan, Ethics, Efficiency, and the Market (Totowa, NJ: Rowman and Littlefield, 1985), p. 29; cited in Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, p. 58 n. 3, and Ch. 5, p. 73.

[2] Formalised from Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4 (pp. 62–68) and Ch. 5 (pp. 77–95); Hahnel, Democratic Economic Planning, Ch. 7 (pp. 130–156); Sandström, Anarchist Accounting (2020), Ch. 3; and Participatory Economy Project, "User Story Mapping" (2023).

[3] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 27–30; Ch. 5, p. 83.

[4] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 3, pp. 40–42; Ch. 5, pp. 81–82.

[5] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 2, pp. 23–25; Ch. 5, pp. 88, 96–97.

- Communities of Affected Parties ($\mathcal{C}^{\mathrm{CAP}} = \{CAP_1, \ldots, CAP_Z\}$): Ecological collectives of residents $\mathrm{pop}(CAP_z) \subseteq \mathcal{I}$ impacted by pollutant $z \in \mathcal{Z}$ in a specific airshed or watershed.[6]
- Six Facilitation Boards ($\mathcal{B} = \{\mathrm{IFB}, \mathrm{CFB}, \mathrm{PFB}, \mathrm{EFB}, \mathrm{HFB}, \mathrm{UFB}\}$), Stand-Alone Capital Asset Entities ($\mathcal{A}^{\mathrm{cap}}$), and the Society Account ($\mathcal{A}^{\mathrm{soc}}$).[7]

Definition 4.1 (Master Catalogue of Seven Priced Economic Categories). Following Albert and Hahnel (1991a, Chs. 5–6), Hahnel (2021, Chs. 7–9), and Szczepanczyk (2023, Sec. 4), the economy tracks supply, demand, and indicative prices across seven disjoint categories indexed by the master catalogue Ω:

$$\Omega = \mathcal{G}^{\mathrm{priv}} \sqcup \mathcal{G}^{\mathrm{pub}} \sqcup \mathcal{G}^{\mathrm{int}} \sqcup \mathcal{R} \sqcup \mathcal{L} \sqcup \mathcal{K} \sqcup \mathcal{Z} \tag{4.1}$$

with complete indicative price vector at iteration t given by:

$$P^{(t)} = \left(p^{\mathrm{priv}}, p^{\mathrm{pub}}, p^{\mathrm{int}}, p^{\mathrm{res}}, p^{\mathrm{lab}}, p^{\mathrm{cap}}, p^{\mathrm{pol}}\right) \in \mathbb{R}_{+}^{|\Omega|} \tag{4.2}$$

4.2 Primal-Dual Reference Model and Sandström's Society Account

In Chapter 5 of The Political Economy of Participatory Economics (pp. 78–79), Albert and Hahnel introduce a primal-dual linear programming reference model to clarify the meaning of shadow prices before presenting their decentralised iterative procedure.[8] Let x be the vector of production activity levels, A be the produced intermediate input matrix, K be the capital stock coefficient matrix (with available stock k^*), R be the non-producible natural resource input matrix (with available ecological supply r^*), L be the direct labour input matrix (with available labour l^*), and v be the row vector of relative social values of net outputs. The economy-wide primal and dual reference problems are:

$$\text{Primal:} \quad \max_{x \geq 0} \; v(I - A)x \quad \text{subject to} \quad Kx \leq k^*, \;\; Rx \leq r^*, \;\; Lx \leq l^* \tag{4.3}$$

$$\text{Dual:} \quad \min_{P_k, P_r, P_l \geq 0} \; P_k k^* + P_r r^* + P_l l^* \quad \text{subject to} \quad P_k K + P_r R + P_l L \geq v(I - A) \tag{4.4}$$

As Albert and Hahnel explicitly note (PEPE Ch. 5, p. 79), the shadow prices P_l solved from the static dual (4.4) reflect only the marginal physical productivities of different labour categories, not the differential disutilities or desirabilities of work from the workers' point of view; whereas the indicative prices generated by Parecon's iterative council procedure incorporate worker well-being and effort disutility directly.[9]

[6] Hahnel, Democratic Economic Planning, Ch. 7 ("The Pollution Demand Revealing Mechanism"), pp. 139–148.

[7] Albert and Hahnel, Looking Forward, Ch. 6 (lf6_1.htm); Sandström, "An Accounting System for a Libertarian Socialist Economy" (2023).

[8] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 78–79.

[9] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, p. 79.

4.2.1 Closing the Macro-Accounting Loop (Sandström 2020/2023)

Anders Sandström's Anarchist Accounting (2020) and 2023 technical specification formalise how opportunity-cost user fees paid by Worker Councils reconcile with effort-based consumer incomes inside the Society Account ($\mathcal{A}^{\text{soc}}$):[10]

1. Social Value of Net Final Output (GDP) at Indicative Prices P:

$$Y^{\text{GDP}}(P) = p^{\text{priv}} \cdot D^{\text{priv}} + p^{\text{pub}} \cdot D^{\text{pub}} + p^{\text{cap,new}} \cdot D^{\text{inv}} \tag{4.5}$$

   In the Society Account, credits consist of all user-right fees charged to Worker Councils for access to society's labour categories ($p^{\text{lab}} \cdot L$), natural resources ($p^{\text{res}} \cdot R$), and productive capital assets held in Stand-Alone Capital Asset Entities ($p^{\text{cap}} \cdot K$), plus Consumer Council depreciation/discount charges on public consumer assets and net WC operating surpluses.

2. Aggregate Consumption Pool, Effort-Weighted Hours (H^{eff}), and Numéraire ($w_{\text{base}} \equiv 1.0$): Let $\sigma_{\text{inv}} \in (0,1)$ be the share of GDP allocated to aggregate investment (DEP Ch. 11),[11] Y^{allow} be the ratified budget for non-working allowances and universal public services (Maxim 4), $Y^{\text{CAP}}(P) = \sum_{z \in \mathcal{Z}} p_z^{\text{pol}} E_z$ be total CAP pollution compensation (funded by WC emission charges $\sum_j p^{\text{pol}} \cdot e_j$), and $\Delta S^{\text{agg}} = \sum_{u \in \mathcal{I}} \Delta S_u$ be net aggregate household borrowing. To ensure exact macro-accounting closure regardless of the distribution of effort ratings ϵ_u, we divide the worker remuneration pool by effective effort-weighted hours $H^{\text{eff}} = \sum_{u \in \mathcal{I}^{\text{work}}} h_u \epsilon_u$ and fix $w_{\text{base}} \equiv 1.0$ as the economy's numéraire:

$$\begin{aligned} Y^{\text{work}}(P) &= (1 - \sigma_{\text{inv}}) Y^{\text{GDP}}(P) - Y^{\text{allow}} - Y^{\text{CAP}}(P) - \Delta S^{\text{agg}} \\ w_{\text{base}}(P) &= \frac{Y^{\text{work}}(P)}{\sum_{u \in \mathcal{I}^{\text{work}}} h_u \epsilon_u} \equiv 1.0 \end{aligned} \tag{4.6}$$

3. Capping Both Mean Workplace Effort Rating and Total Credited Hours (DEP Ch. 6, p. 117): To prevent either collective "grade inflation" or phantom-hours inflation (INV-4B), each worker council WC_j's mean effort multiplier and total credited effort-hours are bounded by its endowed labour $\sum_\ell l_{j,\ell}$ and its social

[10] Sandström, Anarchist Accounting (2020), Ch. 3; Sandström, "An Accounting System for a Libertarian Socialist Economy" (2023), Diagram 1 (Arrows 1–14).

[11] Hahnel and Kerkhoff, "Aggregate Investment Planning," Ch. 11 in Hahnel, Democratic Economic Planning, pp. 219–237.

benefit-to-cost ratio:[12]

$$\frac{\sum_{u\in WC_j} h_u \epsilon_u}{\sum_{u\in WC_j} h_u} \le \min\left(\bar{\epsilon}_{\max},\ \frac{SB_j(P)}{SC_j(P)}\right)$$
$$\sum_{u\in WC_j} h_u \epsilon_u \le \left(\sum_{\ell\in\mathcal{L}} l_{j,\ell}\right) \min\left(\bar{\epsilon}_{\max},\ \frac{SB_j(P)}{SC_j(P)}\right) \tag{4.7}$$

4. Individual Warranted Annual Consumption Budget (B_u): Combining Sandström's four income flows (Sandström 2023, Arrows 1–4), citizen u's warranted budget is:

$$B_u(P) = \underbrace{w_{\text{base}}(P) h_u \epsilon_u}_{\text{1: Work Effort}} + \underbrace{A_u^{\text{social}}}_{\text{2: Benefits}} + \underbrace{\sum_{z:u\in CAP_z} \frac{p_z^{\text{pol}} E_z}{|\text{pop}(CAP_z)|}}_{\text{3: CAP Compensation}} + \underbrace{\Delta S_u}_{\text{4: Loan/Need}}$$
$$\sum_{u\in\mathcal{I}} B_u(P) \equiv (1-\sigma_{\text{inv}}) Y^{\text{GDP}}(P) \tag{4.8}$$

4.3 Formal Optimisation Problems of Councils and CAPs

4.3.1 Consumer Councils and Nested Federations (PEPE Ch. 5, pp. 79–82, 88)

Each consumer council $CC_h \in \mathcal{H}$ seeks to maximise its members' total economic well-being $W_h(U_h, \Delta C_h)$—incorporating both immediate preference fulfilment U_h and endogenous human characteristic development ΔC_h—subject to the equitable budget constraint that its total social cost $SC(CC_h, P)$ does not exceed its members' warranted budget $B(CC_h, P) = \sum_{u\in \text{pop}(CC_h)} B_u(P)$:[13]

$$\max_{y_h^{\text{priv}}, y_h^{\text{pub,loc}} \ge 0} \quad W_h\Big(U_h(y_h^{\text{priv}}, y^{\text{pub}}),\ \Delta C_h(y_h^{\text{priv}}, y^{\text{pub}})\Big) \tag{4.9}$$
$$\text{subject to} \quad SC(CC_h, P) \triangleq p^{\text{priv}} \cdot y_h^{\text{priv}} + p^{\text{pub}} \cdot y_h^{\text{pub,loc}} + |\text{pop}(CC_h)| \sum_{f \succ CC_h} \frac{p^{\text{pub}} \cdot y_f^{\text{pub}}}{|\text{pop}(f)|} \le B(CC_h, P)$$

where $f \succ CC_h$ denotes every higher-level ward, city, state, or national consumer federation containing CC_h. As proven in PEPE Ch. 3 (p. 41) and Ch. 5 (p. 80), for any individual or council whose request satisfies $SC \le B$, other councils may advise but cannot veto the request; veto power applies only to exception requests exceeding warranted allowances ($SC > B$).

[12] Hahnel, Democratic Economic Planning, Ch. 6 ("Capping average effort ratings"), p. 117. To prevent SB_j/SC_j caps from drifting into enterprise output pay (R-ECON-09), default $\bar{\epsilon}_{\max} = 1.0$ applies unless physical overtime/intensity is verified by IF_k, while vintage-equipment plants are equalised via lower $p^{\text{cap}} \cdot k_j^0$ rental charges.

[13] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 77–80, 88.

4.3.2 Communities of Affected Parties (CAPs) (DEP Ch. 7, pp. 139–148)

For each regulated pollutant $z \in \mathcal{Z}$ with hard biophysical sustainability ceiling $E_z^{\max}$ (established by Long-Run Environmental Planning; DEP Ch. 14),[14] given indicative damage price p_z^{pol}, CAP_z chooses the total permissive emission supply $E_z \in [0, E_z^{\max}]$ to maximise member well-being:

$$\max_{0 \le E_z \le E_z^{\max}} \quad W_{CAP_z}\Big(p_z^{\text{pol}} E_z,\ -D_{CAP_z}(E_z)\Big) \tag{4.10}$$

At an interior optimum below $E_z^{\max}$, CAP_z sets E_z such that the sum of marginal victim damages across $u \in \text{pop}(CAP_z)$ equals p_z^{pol} (the Lindahl-Pigou efficiency condition).

4.3.3 Worker Councils (PEPE Ch. 5, pp. 85–88; DEP Ch. 7, pp. 152–156)

Each worker council $WC_j \in \mathcal{J}$ begins the planning cycle with initial labour endowment l_j^0 ($\sum_{j=1}^{J} l_j^0 = l^*$) and initial capital stock endowment k_j^0 ($\sum_{j=1}^{J} k_j^0 = k^*$), for which it is automatically charged $p^{\text{lab}} \cdot l_j^0 + p^{\text{cap}} \cdot k_j^0$.[15] In iteration t, WC_j proposes output vector $\bar{a}_j = (\bar{a}_j^{\text{priv}}, \bar{a}_j^{\text{pub}}, \bar{a}_j^{\text{int}}, \bar{a}_j^{\text{inv}})$, intermediate inputs $\underline{a}_j$, natural resources r_j, net labour additions/releases Δl_j, net capital additions/releases Δk_j, and pollutant emissions e_j from its feasible technology set $\mathcal{T}_j$. Defining Social Benefit (SB_j) and Social Cost (SC_j):

$$SB_j(P) = p^{\text{priv}} \cdot \bar{a}_j^{\text{priv}} + p^{\text{pub}} \cdot \bar{a}_j^{\text{pub}} + p^{\text{int}} \cdot \bar{a}_j^{\text{int}} + p^{\text{cap,new}} \cdot \bar{a}_j^{\text{inv}} \tag{4.11}$$

$$SC_j(P) = p^{\text{int}} \cdot \underline{a}_j + p^{\text{res}} \cdot r_j + p^{\text{lab}} \cdot (l_j^0 + \Delta l_j) + p^{\text{cap}} \cdot (k_j^0 + \Delta k_j) + p^{\text{pol}} \cdot e_j \tag{4.12}$$

WC_j maximises its workers' fulfilment and development $W_j(U_j, \Delta C_j)$ subject to the Socially Non-Abusive Production Constraint (PEPE Ch. 5, pp. 87–88):

$$\max_{\pi_j \in \mathcal{T}_j} \ W_j\Big(U_j(\pi_j),\ \Delta C_j(\pi_j)\Big) \quad \text{subject to} \quad SB_j(P) - SC_j(P) \ge 0 \quad \left(\text{i.e., } \frac{SB_j(P)}{SC_j(P)} \ge 1\right) \tag{4.13}$$

4.4 Critical Algorithmic Audit of pequod-cljs and Convergence Rules

In Chapter 9 of Democratic Economic Planning (2021, pp. 173–194) and Szczepanczyk (2023),[16] the authors simulated 40 experiments (dep1ex61..100.csv, gen-

[14] Hahnel, Democratic Economic Planning, Ch. 7, pp. 139–148, Appendix pp. 159–162, and Ch. 14, pp. 261–267.

[15] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 5, pp. 85–87.

[16] Hahnel, Weisdorf, and Szczepanczyk, "Computer Simulations of Participatory Planning," Ch. 9 in Hahnel, Democratic Economic Planning, pp. 173–194; Szczepanczyk, "Pseudocode and Algorithms" (2023), pp. 43–54; audited against upstream msszczep/pequod-cljs at commit 8bd46b5c233f0e48c767ad6d5069b9437a67c38b (src/clj/pequod_cljs/csvgen.clj, 836 lines; trimmed copy in research/sources/pequod_cljs_csvgen_apache2.clj, 799 lines).

erated via bin/bigger_gen.sh) with 30,000 Worker Councils, 30,000 Consumer Councils, and 500 categories (100 private, 100 public, 100 intermediate, 100 natural resources, 100 labour categories), alongside 50 smaller 3,000-council runs (dep1ex01..50.csv).

4.4.1 The Hahnel-Weisdorf-Szczepanczyk Price Rule and CSV Audit

In csvgen.clj (upstream lines 408–496; local lines 383–468), let $S_i^{(t)}$ be total supply, $D_i^{(t)}$ be total demand, and $\text{surplus}_i^{(t)} = S_i^{(t)} - D_i^{(t)}$. Defining the two-sided relative imbalance $\rho_i^{(t)}$:

$$\rho_i^{(t)} = \frac{2\left|S_i^{(t)} - D_i^{(t)}\right|}{S_i^{(t)} + D_i^{(t)}}, \qquad \tilde{\delta}_i^{(t)} = 1.05 - 0.5^{\rho_i^{(t)}} \tag{4.14}$$

the damped percentage price delta $\delta_i^{(t)}$ and multiplicative price update are:[17]

$$\begin{aligned} \delta_i^{(t)} &= \max\left(0.001,\ \min\left(\tilde{\delta}_i^{(t)},\ \tilde{\delta}_i^{(t)} \cdot \min\left(0.25, \rho_i^{(t-1)}\right)\right)\right) \\ P_i^{(t+1)} &= P_i^{(t)}\left(1 + \text{sign}\left(D_i^{(t)} - S_i^{(t)}\right)\delta_i^{(t)}\right) \end{aligned} \tag{4.15}$$

Our programmatic audit of all 99 committed CSV files (research/analysis/verify_pequod_datasets.py, accounting for the exponent-sum separator rows between Year 1 and Year 2) confirms that across the 40 large-scale runs (dep1ex61..100.csv), Year 1 reaches the 5% threshold (:green) in 11.85 rounds on average and the 3% threshold (:blue) in 19.20 rounds (matching Hahnel 2021, Tables 9.1–9.2, pp. 178–179), while Year 2 reaches 5% in 6.53–6.58 rounds (matching Szczepanczyk 2023, p. 51). Because no wall-clock timing log exists in the upstream repository, any per-iteration wall-clock timing figure (~ 2.5 s) is treated as an unverified estimate.

4.4.2 Audit of pequod-cljs and Proof of "Toothache" Unboundedness (R-ECON-06)

Direct inspection of wcs.clj, ccs.clj, csvgen.clj, and all 150,000 Worker Council records across the 70 dep1ex*.clj datasets establishes four structural properties of pequod-cljs:

1. Unconstrained Closed-Form Cobb-Douglas Optimisation: Each WC_j maximises $\Pi(x,e) = \lambda a e^c \prod_{m=1}^{n} x_m^{b_m} - \sum_{m=1}^{n} p_m x_m - s e^k$ via symbolic Mathematica first-order conditions (solution-1..8, upstream lines 221–401; local lines 209–376) without enforcing $SB_j/SC_j \geq 1$ or labour stock conservation $\sum_j \Delta l_j = 0$.

[17] Szczepanczyk, "Pseudocode and Algorithms" (2023), Eq. 9, p. 48; upstream csvgen.clj lines 408–410 and 480–491.

2. Why "Increasing Returns to Scale" (create-toothache) Violates Concavity and Produces Perverse Saddle-Point Supply (R-ECON-06): In wcs.clj (lines 31–70), each WC_j has $n \in \{3, \ldots, 8\}$ inputs with baseline exponents $\sum_{m=1}^{n} b_m \in [0.75, 0.85]$, effort elasticity $c \in [0.05, 0.10]$, and effort disutility exponent $k \in [3, 4]$ (:du). Eliminating e shows that $\Pi(x, e)$ is strictly concave if and only if $\sum_{m=1}^{n} b_m + \frac{c}{k} < 1$. While one might assume that $k \in [3, 4]$ keeps $\sum b_m + \frac{c}{k} < 1$ under create-toothache, auditing all 150,000 WC records in dep1ex01..70.clj (verify_pequod_datasets.py) proves the opposite:

$$\text{Baseline:} \quad \sum_{m=1}^{n} b_m + \frac{c}{k} \in [0.7706,\ 0.8756] < 1$$

$$\text{Toothache } (+0.1 \text{ per input}): \quad \sum_{m=1}^{n} b_m^{\text{tooth}} + \frac{c}{k} \in [1.0712,\ 1.6652] > 1 \tag{4.16}$$

   Concavity fails for 100% of toothache records (150,000 / 150,000). Along the scaling ray $x_m(t) = t, e(t) = t^{1/k}$ as $t \to \infty$, revenue grows as $t^{\sum b_m + c/k}$ (exponent > 1) against linear cost t^1, so $\sup \Pi = +\infty$! Consequently, the closed-form formula in solution-1..8 evaluates a saddle point (local minimum along the scale ray) where the denominator $k(\sum b_m + c/k - 1) > 0$ flips sign, causing toothache councils to exhibit downward-sloping supply ($\frac{\partial \ln y^{\text{FOC}}}{\partial \ln \lambda} < 0$)—reducing output when output price rises! Thus, pequod-cljs did not validly test non-convex increasing returns.

3. Exogenous Consumer Budgets: In consume (upstream lines 188–204; local lines 176–192), each Consumer Council's budget (:income cc) is fixed at 5,000 across all iterations.

4. Two-Sided Resource Tolerance Without Pollutants (R-ECON-13): Natural resources are given a fixed supply of 1,000 (upstream line 168; local line 162) and priced within the two-sided $\tau = 5\%$ tolerance (permitting up to $\sim 5.1\%$ overshoot), without CAP pollutant pricing or strict one-sided ceiling enforcement.

4.4.3 Why General Convergence Cannot Be Guaranteed by Heuristic Tâtonnement (R-ECON-01–04)

Simply adding heuristic proximal regularisation and trust regions to Eq. (4.15) cannot guarantee convergence under Leontief complementarities and non-convexities, on both theoretical and empirical grounds:

- Scarf's (1960) Leontief Instability & Trust-Region Stalling (R-ECON-03): By the Sonnenschein–Mantel–Debreu theorem and Herbert Scarf's (1960) counterexample,[18] price tâtonnement can cycle indefinitely when income effects

[18] Herbert Scarf, "Some Examples of Global Instability of the Competitive Equilibrium," International Economic Review 1, no. 3 (1960): 157–172; Hugo Sonnenschein, "Do Walras' Identity and Continuity Characterize the Class of Community Excess Demand Functions?" Journal of Economic Theory 6, no. 4 (1973): 345–354.

violate gross substitutability. Running Eq. (4.15) on Scarf's 3-good Leontief economy (verify_economics.py) confirms that over 3,000 rounds, $\max_i \rho_i^{(t)}$ never drops below 10.54% ($\beta \in \{\text{None}, 0.25, 0.10, 0.02\}$), while a decaying trust region ($\beta^{(t)} = 0.25 \cdot 0.99^t$) freezes proposals at a 56.60% imbalance—demonstrating that shrinking trust regions cause stalling, not market clearing!

- Indivisible Plants vs. Shapley–Folkman Smoothing (R-ECON-04): For a single indivisible facility with $S(p) = 100 \cdot \mathbb{I}_{p \geq 10}$ and demand $D(p) = 70 - p$, no clearing price exists: the theoretical minimum imbalance is $\rho = 50.0\%$, and over 500 rounds Eq. (4.15) never falls below 50.37%. However, by the Shapley–Folkman–Starr theorem (Starr 1969),[19] disaggregating that 100-unit capacity across $K = 20$ smaller councils with staggered thresholds in $[8.5, 11.5]$ shrinks the minimum imbalance to 0.40% ($K = 100 \implies \mathbf{0.48}\%$).

4.4.4 Two-Tier Hierarchical MILP + Convex Proximal-Jacobi ADMM (Research Candidate)

To distinguish heuristic smoothing from provable convex decomposition (R-ECON-02), DPEP separates three candidate coordination algorithms to be benchmarked in Phase 1:

1. Algorithm 3a (Proximal Damped Tâtonnement): Adds a non-decaying quadratic proximal penalty $\frac{\mu}{2}\|\pi_c - \pi_c^{(t-1)}\|_2^2$ to smooth council corner jumps while updating prices via Eq. (4.15) under numéraire $w_{\text{base}} \equiv 1.0$.

2. Algorithm 3b (Convex Proximal-Jacobi ADMM): Formulates the continuous clearing problem over convex/Leontief sets $\mathcal{T}_c$ as $\min_{\{\pi_c \in \mathcal{T}_c\}} -\sum_c W_c(\pi_c)$ subject to aggregate clearing $\sum_c A_c \pi_c \leq b$. In round t, each council minimises its Augmented Lagrangian linearization (incorporating the coupled excess-demand gradient $g_{\text{pen}}^{(t)} = \rho_{\text{pen}} A_c^\top E D^{(t-1)}$ alongside prices $P^{(t)}$) with proximal weight $\mu_c > \rho_{\text{pen}} N_{\text{active}} \|A_c\|_2^2$, and the IFB updates dual prices linearly via $P^{(t+1)} = [P^{(t)} + \rho_{\text{pen}} E D^{(t)}]_+$ normalised to $w_{\text{base}} \equiv 1.0$. For closed, proper, convex sets $\mathcal{T}_c$, Proximal-Jacobi ADMM has a mathematically proven $\mathcal{O}(1/t)$ convergence rate to primal feasibility and optimality.[20]

3. Algorithm 3c (Two-Tier Hierarchical Federation MILP + Council ADMM): Because neither tâtonnement nor convex ADMM can clear large indivisible facilities ($K = 1$), Tier 1 (Industry & Regional Federations) resolves discrete binary plant/shift commitments $z_j \in \{0, 1\}$ via Mixed-Integer Linear Programming (HiGHS MILP) at milestone rounds, after which Tier 2 runs Convex

[19] Ross M. Starr, "Quasi-Equilibria in Markets with Non-Convex Preferences," Econometrica 37, no. 1 (1969): 25–38.

[20] Stephen Boyd et al., "Distributed Optimisation and Statistical Learning via the Alternating Direction Method of Multipliers," Foundations and Trends in Machine Learning 3, no. 1 (2011): 1–122; Wei Deng, Ming-Jun Lai, Zhimin Peng, and Wotao Yin, "Parallel Multi-Block ADMM with $o(1/k)$ Convergence," Journal of Scientific Computing 71 (2017): 712–736.

Proximal-Jacobi ADMM across the continuous operating bands of all councils, halting only when the strict one-sided feasibility conditions ($D_g \leq S_g$ for produced goods; $D_r \leq r^*$ and $D_z \leq \min(E_z, E_z^{\max})$ for resources and emissions) are satisfied.

4.5 Protocol State Machine and Canonical Protobuf Schemas

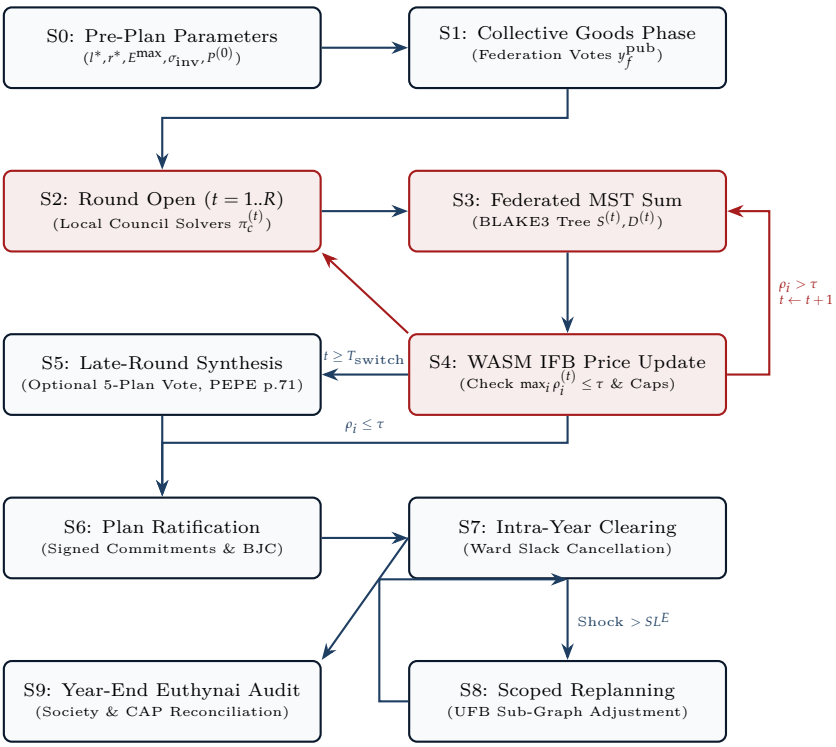


Figure 4.1: Complete 10-State DPEP Annual Planning and Intra-Year Operational State Machine

To prevent IEEE-754 floating-point non-determinism across CPU architectures (x86_64, ARMv8 smartphones, and WebAssembly), all quantities and prices in DPEP are encoded as deterministic 64-bit fixed-point integers (FixedDecimal, default scale 10^{-6} micro-units):

Canonical Wire Encoding (R-SEC-08): Because standard proto3 wire encoding is non-canonical and a variable scale field permits multiple representations of the same number, all signed payloads in DPEP are serialized using RFC 8949 Section 4.2 Canonical CBOR with a fixed global scale of 10^{-6} (int64 micro-units), 128-bit intermediate round-half-to-even multiplication, and degree-6 minimax polynomial fixed-point evaluation for 0.5^{ρ_i}:

```
syntax = "proto3";
package dpep.v1;

message SparseItemVector {
  repeated uint32 item_ids = 1;          // Strictly ascending catalog indices in Omega
  repeated int64 micro_quantities = 2;   // Strictly positive quantities (> 0) in 10^-6 micro-units
}

message QualitativeAnnotation {          // Implements PEPE Ch. 6 'Lists' with GDPR redactability
  uint32 item_id = 1;
  string summary_code = 2;
  string narrative_text = 3;
  bytes redactable_pointer_cid = 4;      // Mutable pointer supporting tombstone erasure (R-TECH-04)
}

message ConsumerCouncilProposal {
  string domain_tag = 1;                 // Fixed: "DPEP-v1-CC-PROPOSAL"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  bytes council_id = 4;
  uint32 accredited_member_count = 5;
  SparseItemVector private_demand = 6;   // Summed from K=3 Prio3 VDAF validated secret shares
  SparseItemVector local_public_demand = 7;
  int64 total_social_cost_micro = 8;
  int64 warranted_budget_micro = 9;
  bool requests_exception = 10;
  repeated QualitativeAnnotation qualitative_list = 11;
  bytes vdaf_transcript_root = 12;       // Commits Prio3 FLP verification transcripts (R-SEC-01)
  bytes signer_bitmap = 13;
  bytes council_hybrid_signature = 14;   // Ed25519 + ML-DSA-65 sig over RFC 8949 Canonical CBOR
}

message WorkerCouncilProposal {
  string domain_tag = 1;                 // Fixed: "DPEP-v1-WC-PROPOSAL"
  uint32 plan_year = 2;
  uint32 round_number = 3;
  bytes council_id = 4;
  uint32 industry_federation_id = 5;
  SparseItemVector proposed_outputs = 6;
  SparseItemVector intermediate_inputs = 7;
  SparseItemVector natural_resources = 8;
  SparseItemVector total_labor_requested = 9;
  SparseItemVector total_capital_requested = 10;
  SparseItemVector pollutant_emissions = 11;
  int64 effective_effort_hours_micro = 12; // Capped by INV-4B: sum(h_u*eps_u) <= L_endowed*min(1,SB/SC)
  int64 social_benefit_micro = 13;
  int64 social_cost_micro = 14;
  int64 sb_sc_ratio_micro = 15;
  bool requests_appeal = 16;               // Subject to Kornai soft-budget cap at IF_k (R-ECON-08)
  repeated QualitativeAnnotation qualitative_list = 17;
  bytes signer_bitmap = 18;
  bytes council_hybrid_signature = 19;
}

message RoundAggregationAndPrices {
  string domain_tag = 1;                   // Fixed: "DPEP-v1-ROUND-CHECKPOINT"
```

```
  uint32 plan_year = 2;
  uint32 round_number = 3;
  repeated int64 indicative_prices_micro = 4; // Normalized so w_base_micro == 1_000_000 (R-ECON-05)
  repeated int64 total_supply_micro = 5;
  repeated int64 total_demand_micro = 6;
  repeated int64 planned_slack_micro = 7;
  repeated int64 one_sided_shortage_micro = 8; // max(0, D_i - S_i) (R-ECON-10)
  int64 base_hourly_remuneration_micro = 9;    // Fixed numeraire: 1_000_000
  bool strict_one_sided_feasible = 10;
  bytes proposals_merkle_root = 11;            // 32-B BLAKE3 root over sorted council proposals (512-B proof)
  bytes category_totals_merkle_root = 12;      // 32-B BLAKE3 root over category sums
  bytes council_participation_bitmap = 13;     // 2 bits/council: 01=Active, 10=Partitioned, 00=Excluded
  bytes deterministic_wasm_code_hash = 14;
  bytes validator_quorum_bitmap = 15;
  repeated bytes validator_hybrid_signatures = 16;
}
```

Listing 4.1: Canonical Schemas for DPEP Proposals and IFB Round Checkpoints (Fixed Micro-Unit Scale 10^{-6})

CHAPTER 5

Distributed Systems Architecture, Consistency, and Low-Power Participation

> "Planning is computerized. All councils can access data banks of all facilitation boards and all other councils. The only thing demanding about flexible information access is that the computer system not only automatically accept but also sum and compare proposals to ascertain the status of any good."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 6, "Preparing First Proposals"[1]

5.1 Why Participatory Economics Is Not a High-TPS Blockchain Problem

When we examine the computational structure formalised in Chapter 4, three architectural truths emerge:

1. Annual Planning Is an Epoch-Synchronised Bulk-Synchronous Parallel (BSP) Process: Millions of participants and tens of thousands of councils draft proposals locally, aggregate them up nested geographic and industrial federation trees, and compute updated indicative prices $P^{(t+1)}$ once per planning round (10–20 rounds per year). Global state finality is required only at iteration boundaries and at plan ratification.
2. Intra-Year Execution Is Hierarchically Scoped: As Albert and Hahnel (1991a, Ch. 4, pp. 66–67) specify, routine intra-year consumer preference shifts cancel out at the neighbourhood, ward, or city federation level within planned slack buffers ($SL^E \approx 3$–5%) without altering aggregate production plans.
3. Participation Must Work Offline and on Resource-Constrained Devices: Workers and consumers in rural collectives or intermittently connected networks must be able to deliberate, optimize local proposals, and sign ballots offline.[2]

5.2 The Four-Layer Hybrid Architecture (Architecture D)

[1] Albert and Hahnel, Looking Forward, Ch. 6 (lf6_2.htm).

[2] Martin Kleppmann, Adam Wiggins, Peter van Hardenberg, and Mark McGranaghan, "Local-First Software: You Own Your Data, in spite of the Cloud," Proceedings of the 2019 ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! '19), pp. 154–178, https://doi.org/10.1145/3359591.3359737.

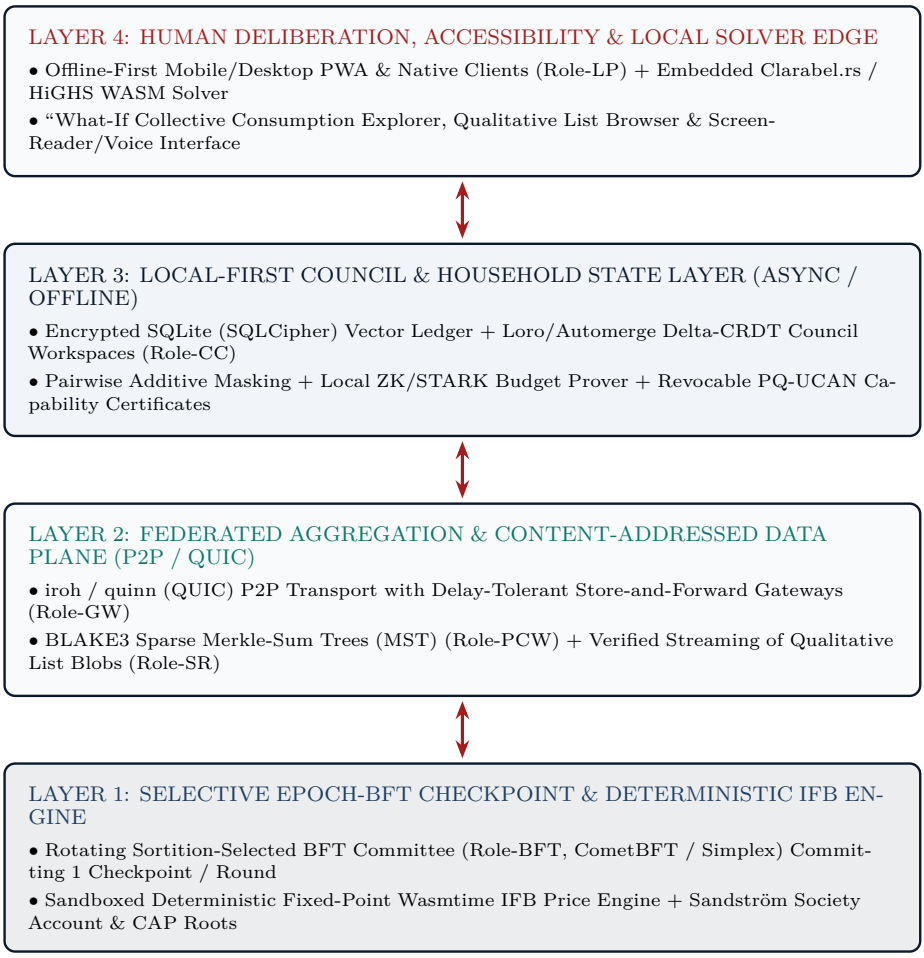


Figure 5.1: Four-Layer Architecture of the Distributed Participatory Economics Protocol (Architecture D)

5.2.1 Two-Root Authenticated Aggregation, Completeness Bitmaps, and Physical Boundaries (R-SEC-06, R-SEC-11)

Naively embedding an 80 KB dense 10,000-category vector inside every internal node of a depth-16 Sparse Merkle-Sum Tree would cause a single mobile inclusion path to bloat to 16×80 KB $= \mathbf{1.28}$ MB—and still could not prevent a dishonest aggregator from fabricating off-path sibling subtrees unless auditors recomputed the entire tree from the leaves up. Moreover, a Merkle inclusion proof for council c proves membership, not completeness (whether dissident council c' was omitted) or physical truth (whether reported output physically exists).

Because all $40,000$ signed sparse council proposals in a round total only $40,000 \times 5.2$ KB $\approx$ **208** MB raw ($\sim$ **60** MB Zstd-compressed), DPEP uses a

Two-Root Authenticated Aggregation Protocol instead of heavy internal-vector MST proofs:

$$H_{\text{proposals}}^{(t)} = \text{BLAKE3-Root}\Big(\big\{\big(\text{id}_c,\ \text{BLAKE3}(\text{CBOR}(\pi_c^{(t)}))\big)\big\}_{c=1}^{M}\Big) \tag{5.1}$$

$$H_{\text{totals}}^{(t)} = \text{BLAKE3-Root}\Big(\big\{\big(i,\ S_i^{(t)},\ D_i^{(t)},\ P_i^{(t+1)}\big)\big\}_{i\in\Omega}\Big) \tag{5.2}$$

- 512-Byte Mobile Inclusion Proofs: Any mobile client (Role-LP) verifies that its council's exact proposal was included in proposals_merkle_root using a standard 16-level hash path of 16×32 B = **512** bytes.
- Full Leaf Data Availability (DA) & Conservation Verification: Every regional Role-SR, Role-PCW, and Role-BFT server downloads the 60 MB compressed leaf archive, verifies all $40,000$ signatures against the accredited council roster, and verifies that their exact fixed-point sum equals category_totals_root before signing the checkpoint.
- Distinguishing Censorship from Network Partitions (council_participation_bitmap): Each checkpoint commits a 2-bit-per-council bitmap (01 = Fresh Round-t Proposal, 10 = Timeout/Partition Carry-Forward, 00 = Excluded). Councils submit proposals to both their primary aggregator and ≥ 2 independent Role-SR relays, receiving timestamped signed SubmissionReceipts. Any council falsely marked 10 or 00 broadcasts its receipt during the pre-finality challenge window $T_{\text{challenge}}$ to force inclusion and fault the censoring aggregator.
- Explicit Cryptographic vs. Physical Boundary (R-SEC-11): Cryptography verifies digital authenticity, inclusion, and arithmetic conservation; it cannot verify that physical grain was harvested or that emissions sensors were not bypassed, which remain governed by federated physical euthynai spot-audits and counterparty delivery reconciliation.

5.3 Granular Consistency Mapping and Node Role Taxonomy

[3] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski, "Conflict-Free Replicated Data Types," Symposium on Self-Stabilizing Systems (SSS 2011), LNCS 6976, pp. 386–400.

[4] Ethan Buchman, Jae Kwon, and Zarko Milosevic, "The Latest Gossip on BFT Consensus," arXiv:1807.04938 (2018); Maofan Yin et al., "HotStuff: BFT Consensus with Linearity and Responsiveness," ACM PODC '19, pp. 347–356.

Table 5.1: Consistency Models and Partition Tradeoffs Across Economic Subsystems

Economic Subsystem	Consistency Model	Latency Target	CAP Partition Behaviour	Architectural Justification
1. Proposal Drafting (S1–S2)	Strong Eventual Consistency (Delta-CRDT + SQLite)[3]	< 50 ms local	AP (Offline-First): Drafts continue on local LAN/device.	Unsubmitted drafts affect no external council; offline usability is paramount.
2. Council Ratification Vote	Council-Scoped Quorum (k-of-n threshold sig)	Seconds to hours	Locally CP on Council LAN: Independent of WAN.	Only members of WC_j or CC_h hold authority over their council's proposal.
3. Federated Round Aggregation (S3)	Deterministic Convergent Tree Reduction (MST)	1–10 min per round	Bounded-Wait CP + Fallback: Carries forward $\pi_c^{(t-1)}$ if WAN severed past grace window.	Prevents a single disconnected hamlet from halting national planning while preserving verifiable inclusion.
4. IFB Price Update & Final Plan (S4, S6)	Global Total Order / Linearizable Epoch BFT[4]	2–15 s per block	CP (Safety over Liveness): Requires $\geq 2f+1$ of $3f+1$ federation validators.	Every council must optimise round $t+1$ against the exact same indicative price vector $P^{(t+1)}$.
5. Intra-Year Fulfilment (S7)	Bilateral Signed Receipts + Federated Rollup	< 500 ms (offline-capable)	AP with Bounded Credit: Daily depot credit cap reconciled at Ward rollup.	Point-of-delivery pickups at neighbourhood depots must never block on global consensus.

Table 5.2: Six-Role Hardware Realism Matrix (Supporting Legacy Smartphones and Raspberry Pis)

Node Role	Core Responsibilities	Entry Smartphone (3GB RAM ARM)?	Raspberry Pi 4/5 (4–8GB RAM)?	Community Server (8+ Cores)?
1. Lightweight Participant (Role-LP)	Local household budget QP solver, ZK budget proof, CRDT sync, MST & BFT header verifier, ballot signer.	YES (Primary) (< 50 MB RAM, $< 1\%$ battery/day)	YES	YES
2. Council Coordinator (Role-CC)	Hosts WC_j or CC_h local Wi-Fi CRDT relay, runs workplace LP/QP solver, assembles council threshold sig.	YES (Failover Hotspot)	YES (Ideal \$50 Appliance at 5W)	YES
3. Storage Replica (Role-SR)	Pins regional content-addressed BLAKE3 blobs for qualitative Lists and audit logs.	NO (Flash wear & cell data limits)	YES (with USB SSD)	YES (Primary)
4. Planning Compute Worker (Role-PCW)	Computes federated MST reductions, deterministic WASM price updates, and input-output analytics.	NO	PARTIAL (Up to $5,000$ councils)	YES (Primary IFB Node)
5. Epoch Consensus Validator (Role-BFT)	Participates in sortition-elected BFT committee finalising round checkpoints.	NO (Requires 24/7 fixed routing)	YES (For PoC / local federations)	YES (Primary)
6. Delay-Tolerant Gateway (Role-GW)	Bridges offline mesh/LoRa/USB sneakernet bundles to federation relays.	YES (As "Data Mule")	YES (Mesh Box)	YES

CHAPTER 6

Weighted Architecture and Implementation Stack Comparison

> "Devising and choosing from among these and other possibilities is a practical issue in implementing any actual participatory economy. Assuming the procedures chosen do not violate principles essential to participatory planning, considerations include (1) the extent to which iteration workers could bias outcomes, (2) the extent of reductions in the number of iterations required to reach a plan…"
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 4, p. 68[1]

6.1 Weighted Comparison of the Four Architectural Paradigms

We compare the four candidate architectures across all 14 required criteria: Architecture A (Conventional Federated SQL/REST Services), Architecture B (Pure P2P Gossip + CRDTs without finality), Architecture C (Full Per-Transaction BFT Blockchain), and Architecture D (Hybrid Local-First + Federated Merkle-Sum Trees + Selective Epoch-BFT Checkpoints).

6.1.1 Sensitivity Analysis of Architecture Rankings (R-TECH-05)

While Architecture D leads under our baseline long-term weights (4.51 vs. 3.59 for P2P, 3.10 for Federated SQL/REST, and 2.54 for a Blockchain) and under equal 1/14 weights (4.46 vs. 3.61, 3.32, 2.46), sensitivity analysis shows that **when 70% of weight is placed on rapid prototyping simplicity, throughput, and low operational complexity (as in Phase 1 and Phase 1H), Architecture A (Conventional Federated Services) outscores Architecture D (4.65 vs. 3.60)**. We therefore use Architecture A for Phase 1H human-factors testing and reserve Architecture D for Phase 2–3 distributed deployment.

---

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 4, p. 68.

[2] Analytical derivation from Listing 4.1 schemas (at $N = 10^6$ participants, $G = 10,000$ categories): each Layer-1 RoundAggregationAndPrices checkpoint block contains 5 FixedDecimal arrays ($5 \times 10,000 \times 8$ B = 400 KB) plus 64 hybrid Ed25519 + ML-DSA-65 validator signatures ($64 \times 3,373$ B $\approx$ 216 KB) $\approx$ 616 KB/block. Over 20 annual planning rounds + 365 daily operational rollups + 50 replanning epochs (435 blocks/year), uncompressed Layer-1 growth is 435×616 KB $\approx$ 268 MB/year ($<$ 1 GB/year), while full Layer-2 council proposal archives (Role-SR) grow by $\sim$ 78 GB/year (Table 9.2). Hosting 100 federation servers at \$30–\$60/month costs \$3,000–\$6,000/month ($\approx$ \$0.003–\$0.006/citizen/month, or \$0.036–\$0.072/citizen/year). These are unbenchmarked analytical projections to be empirically tested against database indexing and P2P retransmission overhead in Chapter 9.

Table 6.1: Weighted 14-Criterion Comparison of Architectures A, B, C, and D (1–5 Scale)

Criterion	Wt.	Arch A (REST)	Arch B (P2P)	Arch C (Chain)	Arch D (Hybrid)	Engineering & Normative Rationale
1. Decentralisation	9%	2.0	5.0	3.0	4.5	Arch A centralizes DB admin power; Arch C centralizes heavy validators; Arch D combines local autonomy with rotating BFT.
2. Trust Assumptions	10%	2.0	3.0	4.5	4.5	Arch D matches blockchain cryptographic verifiability via MST inclusion proofs + deterministic WASM at 100× lower cost.
3. Censorship Resistance	8%	2.0	4.5	4.0	4.5	Signed submission receipts allow any censored council to prove omission directly to the Epoch BFT committee.
4. Throughput & Latency	8%	5.0	4.0	2.0	5.0	Daily operations clear locally in < 10 ms; 100k council proposals aggregate in parallel in < 5 s.
5. Network Bandwidth	7%	4.5	2.5	1.5	4.5	Tree reduction + Zstd price deltas avoid $\mathcal{O}(N^2)$ gossip/block flooding.
6. Hardware Requirements	8%	4.5	3.5	1.5	4.5	Runs on $50 Raspberry Pi (Role-CC) and legacy Android (Role-LP).
7. Offline Capability	9%	1.5	5.0	1.0	5.0	Native SQLite + Loro/Automerge CRDTs + DTN store-and-forward.
8. Byzantine Fault Tolerance	8%	1.0	2.5	5.0	4.5	MST conservation checks + Epoch BFT prevent equivocation and phantom supply/demand injection.
9. Privacy	9%	2.5	2.0	1.0	4.5	Public ledgers (Arch C) leak personal consumption; Arch D uses pairwise masking + local ZK budget proofs.
10. Post-Quantum Readiness	6%	4.0	3.5	2.0	4.5	Edge MST verification absorbs 3.3 KB ML-DSA-65 signatures without bloating global blocks.
11. Implementation Simplicity	6%	5.0	3.0	2.5	3.5	Modular 4-layer stack allows clean phased rollout.
12. Operational Cost	4%	4.0	4.5	2.0	4.5	Analytical projection: < 1 GB/yr L1 checkpoint growth (~ 78 GB/yr L2 archive); $\sim$ $0.005/citizen/mo ($\sim$ $0.05/yr) server hosting.[2]
13. Energy Efficiency	4%	4.5	4.0	2.5	4.5	Epoch-driven consensus uses $\sim 10^{-7}\times$ Proof-of-Work energy.
14. Maintainability	4%	4.0	3.5	2.0	4.0	Economic rules compiled to upgradable WASM modules.
WEIGHTED SCORE	100%	3.10	3.59	2.54	4.51	Arch D leads long-term; Arch A leads Phase 1.

6.2 Two-Stage Language Strategy: Python-First Research to Rust Protocol

Rather than locking the project prematurely into systems-level Rust before the non-convex economic coordination algorithms are validated, we adopt a two-stage implementation strategy:

1. Phase 1 (Economic Coordination Research) — Python Scientific Stack (NumPy/SciPy, highspy MILP/LP, Clarabel Python bindings): Minimizes iteration time and maximises reproducibility for testing non-convex ADMM, MILP federation decomposition, and Scarf/Leontief stress tests.
2. Phase 2 & 3 (Validated Distributed Protocol & Mobile Engine) — Rust + WebAssembly (dpep-core, dpep-node): Translates the Phase-1-validated clearing rules into memory-safe Rust (Clarabel.rs,[3] RustCrypto FIPS 203/204/205, quinn QUIC, Wasmtime, and Mozilla UniFFI). While local council solvers (Clarabel/HiGHS) use IEEE-754 floating-point heuristics to draft proposals, all protocol-level verification ($SB_j/SC_j \geq 1, SC_h \leq B_h$, and IFB price updates) executes in deterministic 64/128-bit fixed-point arithmetic (R-TECH-03).

Table 6.2: Programming Language Comparison Across Core Protocol Requirements (1–5 Scale)

Technical Dimension	Rust	Go	TypeScript	Kotlin	Swift	WASM (from Rust)
Memory Safety & Zero GC Pauses	5.0	3.5	2.5	3.0	4.0	5.0 (Sandboxed linear memory)
NIST PQC (ml-kem, ml-dsa, slh-dsa)	5.0	4.0	2.0	2.5	2.5	5.0 (Direct compilation)
Local-First & P2P (iroh, CRDT, SQLite)	5.0	4.5	4.0	3.0	3.0	4.5 (Browser worker target)
Operations Research (Clarabel.rs, HiGHS)	4.5	2.5	2.0	2.5	2.5	5.0 (Runs on every phone/PC)
Cross-Platform Mobile Reuse (UniFFI)	5.0	3.0	4.5	3.0	1.5	5.0 (Universal execution)
Accessible UI & Screen-Reader Tooling	2.5	1.5	5.0	4.5	4.5	Paired with TypeScript PWA UI

[3] Paul J. Goulart and Yuwen Chen, "Clarabel: An Interior-Point Solver for Conic Programs with Quadratic Objectives," University of Oxford (2024), https://github.com/oxfordcontrol/Clarabel.rs.

CHAPTER 7

Post-Quantum Cryptography, Democratic Membership, and Privacy

> "Decisions about what individuals wish to consume will be subject to collective criticism by fellow council members, though with specific guarantees for preserving individual freedoms and privacy."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 3, "Consumption Norms"[1]

7.1 Standardised Post-Quantum Cryptography (NIST FIPS 203, 204, 205)

To protect multi-decade household consumption and health-need records against "harvest-now, decrypt-later" quantum adversaries while avoiding unsubstantiated claims of unconditional lattice security, DPEP employs cryptographic agility (CryptoSuiteId) and uniform NIST Security Level 3 hybrid classical + post-quantum constructions based on the NIST standards finalised in August 2024:[2]

7.2 Democratic Membership, Sybil Resistance, and Civil Liberties (R-SEC-07, R-POL-06)

Design Principle 7.1 (Four-Way Separation of Identity, Eligibility, Accountability, and Legitimacy). DPEP strictly separates: (1) Cryptographic Identity (control of a keypair), (2) Institutional Eligibility (accredited membership in a council), (3) Economic Accountability (correspondence of reported claims to physical reality), and (4) Democratic Legitimacy (self-management proportional to degree affected).

While phantom digital accounts in CC_h cannot mint adult work income without WC_j effort certificates capped by SB_j/SC_j, four deeper Sybil and collusion vectors

[1] Albert and Hahnel, Looking Forward, Ch. 3 (lf3_2.htm); Michael Albert, Parecon: Life After Capitalism (Verso, 2003), Ch. 12 ("Consuming") and Ch. 18 ("Privacy / Frenzy: A Busybody Economy?"), where Albert specifies that individual consumption proposals within warranted effort-rated budgets are submitted anonymously to prevent "busybody" intrusion; see also Jan Philipp Dapprich, "Democratic Theory and Economic Planning: A Critique of Participatory Economics," Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024), https://ejpe.org/journal/article/view/867.

[2] National Institute of Standards and Technology (NIST), FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) (August 13, 2024). Note that while byte sizes in Table 7.1 are exact specification constants, ARMv8 latencies are analytical order-of-magnitude estimates to be empirically benchmarked on target legacy devices in Phase 3 (R-SEC-10).

Table 7.1: Post-Quantum Cryptographic Primitives, Byte Sizes, and Analytical ARMv8 Latency Estimates

Primitive & Standard	Security	Pub Key	Cipher/Sig	ARM Sign	ARM Verify	Assigned Role in DPEP
X25519 + ML-KEM-768 (FIPS 203 Hybrid KEM)	NIST L3 + Classical	1,216 B	1,120 B	$\sim 80\ \mu s$	$\sim 85\ \mu s$	Point-to-point QUIC channels & Prio3 aggregator share encryption.
Ed25519 + ML-DSA-65 (FIPS 204 Hybrid Sig)	NIST L3 + Classical	1,984 B	3,373 B	$\sim 230\ \mu s$	$\sim 175\ \mu s$	Workhorse Signature for council proposals, PQ-UCAN tokens & receipts.
SLH-DSA-SHA2-192s (FIPS 205 Hash-Based)	NIST L3 (Hash Only)	48 B	16,224 B	~ 250 ms	~ 4.5 ms	Level-3 Root Anchor for Federation accreditation & WASM rule signing (R-SEC-10).
BLAKE3-256 / SHA3-256 (FIPS 202 Option)	256b Preimage / 128b Coll.	N/A	32 B	$< 1\ \mu s$/KB	$< 1\ \mu s$/KB	Two-Root Merkle trees & content CIDs ($\sim$ 128b Grover preimage / $\sim$ 85b BHT collision).

must be closed explicitly (R-SEC-07):

1. Closing Workplace Hours Inflation (INV-4B): Capping only the mean effort rating $\bar{\epsilon}_j$ fails if a WC_j fabricates phantom workers to inflate total hours $\sum h_u$. Equation (4.7) closes this by capping total credited effort-hours $\sum_{u \in WC_j} h_u \epsilon_u \leq (\sum_\ell l_{j,\ell}) \min(\bar{\epsilon}_{\max}, SB_j / SC_j)$ and flooring labour user-right charges at $p_\ell^{\text{lab}} \geq w_{\text{base}}$.
2. Detecting Collusive "Carousel Fraud" Between Fake Producer and Consumer Councils: If a colluding WC_j and CC_h sign fictitious delivery receipts for non-existent services to mint effort credits, cryptography alone cannot detect the lie (R-SEC-11). DPEP pairs automated input-output ratio anomaly flags (detecting closed-loop service claims without corresponding physical/intermediate inputs) with mandatory Industry Federation physical euthynai spot-audits prior to accrediting capacity.
3. Preventing Vote & Sortition Dilution: Sortition tickets and federation referenda are weighted by cross-audited council accreditation quotas rather than raw public keys.
4. Civil-Liberties Protection Against a Coercive Population Register (R-POL-06): To prevent housing (HFB) and workplace (EFB) registries from forming a surveillance state database or trapping victims of domestic coercion, citizens convert their council attestations into unlinkable blinded epoch budget credentials (unlinking residency from workplace and consumption), wallets are strictly individual rather than household-head controlled, and unaffiliated migrants/transients receive unconditional Maxim-4 baseline provisioning via community clinics.

7.3 Validated Private Household Aggregation via Prio3 VDAFs (R-SEC-01–05)

A naive privacy design in which each household u generates a standalone zero-knowledge budget proof Π_u on a commitment C_u while separately submitting a 1-round pairwise-masked vector $\tilde{y}_u$ suffers from three severe flaws: (1) Unbound Inputs (R-SEC-01): nothing binds C_u to $\tilde{y}_u$, allowing a user to prove compliance for $y_u = \mathbf{0}$ while injecting huge demand into $\tilde{y}_u$; (2) Offline Dropout Fragility (R-SEC-02, R-SEC-03): if even 1 of 100 neighbors drops offline, pairwise masks fail to cancel, and establishing 99 fresh ML-KEM-768 seeds + dense 10k vectors costs $\sim$ 191 KB; and (3) Proximity-Gap Fragility (R-SEC-05): Crites and Stewart (IACR ePrint 2025/2046) disproved the up-to-capacity Reed–Solomon correlated-agreement conjectures underlying aggressive FRI/STIR/WHIR parameterizations,[3] requiring conservative Johnson-bound FRI proofs ($\sim$ 65–110 KB)

[3] Elizabeth Crites and Alistair Stewart, "On Reed–Solomon Proximity Gaps Conjectures," IACR Cryptology ePrint Archive 2025/2046 (December 2025), https://eprint.iacr.org/2025/2046.pdf.

wherever STARKs are used.

DPEP Solution — IETF CFRG Prio3 / VDAF (draft-irtf-cfrg-vdaf-16):[4] Each household u splits its private demand vector y_u^{priv} into $K = 3$ additive finite-field secret shares $y_u^{\text{priv}} = y_{u,1} + y_{u,2} + y_{u,3} \pmod q$ encrypted via X25519MLKEM768 to $K = 3$ institutionally separate aggregators (Neighbourhood CC_h, Ward Federation, and Consumer Federation), accompanied by a Fully Linear Proof (FLP) verifying directly on the secret shares that:

$$\forall g \in \mathcal{G}^{\text{priv}} : \; y_{u,g}^{\text{priv}} \in [0, Y_{\max}] \quad \wedge \quad \sum_{g \in \mathcal{G}^{\text{priv}}} p_g^{\text{priv}} y_{u,g}^{\text{priv}} + \text{PubShare}_u \le B_u \tag{7.1}$$

Properties of Prio3 VDAF in DPEP: (a) 100% Binding: the exact shares verified by the FLP are the shares summed into y_h^{priv}; (b) 100% Offline Dropout Resilient: each household's submission is verified and summed independently in a single pass; (c) 100% Post-Quantum: relies solely on finite-field secret sharing, SHAKE256/AES PRGs, and ML-KEM-768 (zero lattice/pairing/Reed–Solomon conjectures); and (d) Residual Leakage Mitigations (R-SEC-04): enforces a minimum batch size $m_{\min} \ge 30$, automatically carries forward prior-round shares for offline members to prevent dropout differencing, aggregates rare/niche goods at the Ward level ($m \ge 500$), and uses blinded B_u credentials inside the circuit.

7.4 Constitutional Governance of Protocol Parameters & Sortition (R-POL-01, R-POL-02, R-SEC-09)

To prevent software maintainers or server-rich councils from becoming a new coordinator class:

1. Dual-Federation Ratification of WASM Rules, Catalogues, and Caps (R-POL-01): Changes to wasm_rule_hash, product categories Ω, or 5-plan voting grids require a $\ge 2/3$ dual supermajority of the National Consumer Federation and Congress of Industry Federations with a 60-day reproducible-build audit window, and cannot be altered mid-plan.
2. Hardware-Decoupled Citizen Sortition (R-POL-02): Sortition draws human citizen delegates uniformly across all accredited councils first (via a commit-reveal hash beacon anchored by outgoing SLH-DSA-192s signatures), and provisions selected citizens with community-hosted Role-BFT validator appliances operated under their personal keys.
3. Stratified $n = 128$ Annual Committee Rotation (R-SEC-09): To avoid the 89.1% annual failure risk of per-block redraws at 20% pool corruption, a strat-

[4] Henry Corrigan-Gibbs and Dan Boneh, "Prio: Private, Robust, and Scalable Computation of Aggregate Statistics," USENIX NSDI '17, pp. 259–282; Dan Boneh et al., "Zero-Knowledge Proofs on Secret-Shared Data via Fully Linear PCPs," CRYPTO 2019; Richard Barnes et al., Verifiable Distributed Aggregation Functions (VDAFs), IETF CFRG draft-irtf-cfrg-vdaf-16 (2025), https://datatracker.ietf.org/doc/html/draft-irtf-cfrg-vdaf-16.

ified committee of $n = 128$ validators serves for the full annual planning cycle (1–4 rotations/year) with forward-secure key erasure upon handoff.

CHAPTER 8

Accountable AI, Mathematical Optimisation, and Failure Modes

> "Finally, we point out to those who fear that computers could become the new dictators, the computer programmes are human products, socially evaluated and improved each year… In short, neither consumers nor producers need accept computer facilitated alternatives."
>
> — Michael Albert and Robin Hahnel, Looking Forward (1991), Ch. 9, "Revising Data in Subsequent Iterations"[1]

8.1 Architectural Rule: Deterministic Core, Advisory Edge

AI must never become an unaccountable central planner or surrogate coordinator class. DPEP enforces a strict boundary:

- Core Clearing & Pricing Engine (100% Non-AI): All price updates, SB/SC ratios, and Society Account settlements use deterministic fixed-point arithmetic and exact Operations Research (OR) solvers (Clarabel.rs / HiGHS) with verifiable Karush-Kuhn-Tucker (KKT) optimality certificates.[2]
- Edge Advisory & Accessibility Support: Classical hierarchical time-series models (ETS/ARIMA) forecast SKU size/color distributions; local QP solvers draft initial proposals from last year's bundle (eliminating David Schweickart's "consumption list burden" objection);[3] and optional local open-weight LLMs are restricted to translation, voice/screen-reader accessibility, and summarising qualitative Lists with mandatory links to raw council text.

8.2 Substantive Engagement with Major Economic & Democratic Critics (R-POL-04, R-ECON-08)

We engage directly with the central theoretical critiques of Participatory Economics across philosophy, economics, and mechanism design—including the Winter 2024 special issue of the Erasmus Journal for Philosophy and Economics (John

[1] Albert and Hahnel, Looking Forward, Ch. 9 (lf9_2.htm).

[2] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein, "Distributed Optimisation and Statistical Learning via the Alternating Direction Method of Multipliers," Foundations and Trends in Machine Learning 3, no. 1 (2011): 1–122.

[3] David Schweickart, Against Capitalism (Cambridge: Cambridge University Press, 1993), pp. 329–334.

Table 8.1: Evaluation of Key Decision-Support Capabilities Across Section 10 Criteria

Capability	Best Method (Is AI Needed?)	Local?	Verifiability & Challengeability	Override & Power Concentration Safeguards
1. SKU Demand Forecasting	No LLM needed: Hierarchical ETS/ARIMA + demographics.	Yes (< 50 ms)	Published MAPE error bounds & open parameters.	Non-binding baseline; councils edit freely.
2. Workplace Input Optimisation	No AI needed: Conic QP / LP (Clarabel.rs / HiGHS).	Yes (< 10 ms in WASM)	Exact KKT dual certificates prove optimality.	Presents 3 Pareto-frontier options for worker council vote; democratizes OR tools.
3. Household Proposal Drafting	No AI needed: Deterministic diff + constrained utility QP.	Yes (100% on phone)	Line-by-line diff against prior year's bundle.	Mandatory review UI highlights top changes; zero data leaves user's device.
4. Tradeoff Explanation	Deterministic Leontief Cost-Push Attribution ($(I-A)^{-1}$).	Yes (< 50 ms)	Exact algebraic decomposition of ΔP_i.	Avoids LLM hallucination by rendering exact resource/labour/pollution cost shares.
5. Qualitative List Translation	Yes: Local Neural Machine Translation + TTS.	Yes (Role-CC)	Original source text & BLAKE3 CID always linked alongside.	Bilingual delegates can post verified overrides; expands cross-regional inclusion.
6. Late-Round 5-Plan Synthesis	No AI needed: Parametric LP over 5 fixed weight vectors.	Yes (< 2 s)	Deterministic WASM LP verifiable by any council.	Councils can vote to reject all 5 plans and continue iterating (LF Ch. 9 lf9_2.htm).

O'Neill, Hannes Kuch, Jan Philipp Dapprich),[4] Seth Ackerman (2014),[5] Paul Cockshott and Allin Cottrell (1993),[6] János Kornai (1980, 1986), Martin Weitzman (1974, 1980), and David Schweickart (1993):

1. Dapprich (2024) on Democratic Theory, Liberty, and Concrete Plans vs. General Procedures (R-POL-04): Dapprich evaluates three justifications for democracy—(1) protection of interests, (2) preference revelation, and (3) liberty—and argues that citizens should democratically choose general algorithmic planning procedures and macro-rules rather than voting on concrete plans or subjecting individual consumption to neighbour vetoes.

    - Where Dapprich Is Right (Personal Consumption Autonomy): Subjecting a household's personal consumption within its warranted budget B_u to neighbour oversight violates personal liberty. That is why DPEP enforces INV-5 and Prio3 secret-shared privacy: personal consumption within B_u is mathematically non-vetoable and invisible to neighbours!
    - Why Algorithmic Centralism Fails for Workplaces & Collective Goods: Unlike atomised retail purchases, how a workplace organises its pace of work, safety, and emissions—or how a city designs transit and parks—is an irreducibly collective, non-separable activity. If workers cannot collectively propose and revise their own workplace's input-output-effort plan and must instead obey outputs computed by a central procedural algorithm, they are reduced to wage employees of an algorithmic auctioneer.

2. Cockshott & Cottrell (1993) on Direct Linear Programming and Scalar Labour-Time (R-POL-04): Cockshott and Cottrell propose replacing iterative council negotiation with direct supercomputer LP solving measured in embodied labour hours. This fails on two grounds: (a) scalar labour-time ignores the independent scarcity of non-producible natural resources (r^*), ecological sinks ($E_z^{\max}$), and fixed capital (k^*) as well as differential workplace disutility; and (b) direct central LP solving requires every workplace to transmit its complete private technology set $\mathcal{T}_j$ to a central authority, violating Hayek's (1945) tacit-knowledge insight and worker self-management. In DPEP, central MILP/LP solvers are used strictly as Phase 1 reference oracles and Tier-1 federation aids for discrete lumpy capacities, never to dictate internal workplace life.

[4] John O'Neill, "Incommensurability, Environment and Planning" and "Socialism, Ecology and Economic Calculation," Erasmus Journal for Philosophy and Economics 17, no. 2 (Winter 2024): 29–58; Hannes Kuch, "Decentralization, Federations, and Authority in a Participatory Economy," EJPE 17, no. 2 (2024); Jan Philipp Dapprich, "Democratic Theory and Economic Planning: A Critique of Participatory Economics," EJPE 17, no. 2 (2024), https://ejpe.org/journal/issue/download/35/67.

[5] Seth Ackerman, "Horizontalism and Idealism in Socialist Imagination: An Appraisal of the Participatory Economy," Science & Society 78, no. 2 (2014): 207–239.

[6] W. Paul Cockshott and Allin Cottrell, Towards a New Socialism (Nottingham: Spokesman Books, 1993).

3. Ackerman (2014), Kornai (1980), and Weitzman (1974, 1980) on Dynamic Entry, Soft Budgets, and the Ratchet Effect (R-ECON-08):

   - Dynamic Firm Entry (Ackerman 2014; Kuch 2024): To prevent Walrasian stasis or incumbent guild gatekeeping, any accredited worker group denied startup accreditation by an Industry Federation can appeal to a Cross-Industry Innovation Jury (selected by sortition and co-ratified by the Consumer Federation) backed by a dedicated R&D capital slice.
   - Hardening Soft Budget Constraints (Kornai 1980, 1986): If councils with $SB_j/SC_j < 1$ could obtain unlimited requests_appeal waivers, budget discipline would collapse.[7] DPEP limits appeal waivers to 1 planning cycle (requiring physical retooling audit) and funds them from a capped Industry Transition Pool so granting a waiver directly reduces peer councils' discretionary industry reserve.
   - Avoiding the Soviet Ratchet Effect (Weitzman 1980): Because a council's endowment charge $p^{\text{cap}} \cdot k_j^0 + p^{\text{lab}} \cdot l_j^0$ is based on its physical asset inventory and workforce roster rather than ratcheted up from last year's voluntary over-production, councils are not penalised in Year $T+1$ for high efficiency in Year T.

4. Ecological Value Incommensurability & Environmental Justice in CAPs (R-POL-07 — O'Neill 2024): To address O'Neill's (2024) incommensurability critique and prevent environmental injustice (where a regional CAP accepts toxic exposure for consumption credits, or acts as a bilateral monopoly), DPEP enforces: (a) strict one-sided biophysical ceilings $(r^*, E_z^{\max})$ set by scientific and intergenerational Environmental Planning (following Weitzman 1974 on quantity caps for steep damage thresholds); (b) a hyper-local plume sub-zone veto for the most-impacted neighbourhood $CC_h \subset CAP_z$; and (c) earmarking $\geq 50\%$ of $p_z^{\text{pol}} E_z$ revenues to collective remediation and preventative health.

5. Governance of Solver Defaults & Deliberative Fatigue (R-POL-03 — Devine 1988; Levy 1991; Albert 2003, Ch. 18): Even without AI, local QP solver defaults could become de facto decisions if citizens rubber-stamp them. DPEP bans silent auto-submission of $> 5\%$ effort or category shifts, requires active choice among ≥ 3 Pareto-frontier options at milestone rounds, and tests actual override rates in Phase 1H.

6. Why Black-Market Arbitrage Cannot Snowball into Wage Labour (R-POL-05 — Albert 2003, Ch. 20): While Prio3 budget proofs keep personal bundles private (so neighbours cannot snoop on personal items), they mathematically prevent spending more planning credits than B_u, and an unauthorised private

[7] János Kornai, Economics of Shortage (Amsterdam: North-Holland, 1980); Kornai, "The Soft Budget Constraint," Kyklos 39, no. 1 (1986): 3–30; Martin L. Weitzman, "Prices vs. Quantities," Review of Economic Studies 41, no. 4 (1974): 477–491; Weitzman, "The 'Ratchet Principle' and Performance Incentives," Bell Journal of Economics 11, no. 1 (1980): 302–308.

firm cannot acquire intermediate inputs $(\underline{a}_j, r_j, k_j)$ through the federated planning tree. (Off-protocol physical barter of already-acquired personal goods remains a social question outside software.)

CHAPTER 9

Three-Phase Experimental Programme, Kill Gates, and Benchmarks

> "The easiest 'experiments' to implement are a variety of computer simulations in which various features of the system and assumptions about actors' behaviours can be altered to see how outcomes change… Conceivably one could run a 'shadow' participatory economy within an existing economy."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Feasibility," pp. 108, 129[1]

9.1 Why DPEP Uses Three Gated Phases Instead of a Single Monolithic Build

Attempting to build non-convex economic solvers, QUIC P2P networking, Merkle aggregation, deterministic WASM execution, hybrid post-quantum cryptography, zero-knowledge proofs, and mobile UIs simultaneously in a single monolithic Proof of Concept would conflate economic mechanism validation with distributed systems engineering. Instead, we structure the programme into three progressively validated experimental phases (plus a parallel low-tech human participation probe, Phase 1H), where Phase 1 acts as a strict scientific kill gate: no distributed systems or cryptographic build should be funded until Phase 1 proves that the economic coordination mechanism converges reliably under realistic industrial non-convexities and ecological caps.

9.2 Multi-Scale Analytical Projections (10^2 to 10^6 Participants)

Methodological Note (Category B Analytical Estimates): Only the single-JVM pequod-cljs convergence count (6.5–19.2 rounds) and our Python stress tests (verify_economics.py) have been empirically executed to date. All distributed figures in Table 9.2 are analytical estimates derived from our canonical CBOR (5.2 KB/council) and Prio3 VDAF schemas, incorporating a $1.5\times$–$3\times$ real-world indexing/retransmission range and the 250–400 KB mobile round budget (R-TECH-01, R-TECH-02).

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 108, 129.

Table 9.1: Staged Three-Phase Experimental Programme, Kill Gates, and Revised Budgets

Phase & Stack	Core Experiments	Mandatory Go / No-Go Kill Gate	Duration, Staff & Cost
Phase 1: Economic Coordination Research (Python, NumPy, HiGHS MILP/LP, Clarabel)	8 Stress Regimes: (1) Convex baseline; (2) Leontief & Scarf cycles; (3) Fixed setup costs ($K = 1..100$); (4) Capacity limits; (5) Endogenous income + numéraire; (6) Intra-year shocks; (7) Strategic appeals; (8) One-sided ecological caps.	KILL GATE 1: $\geq 95\%$ convergence to strict one-sided feasibility ($D_g \leq S_g, D_r \leq r^*$) in ≤ 25 rounds on 1,000-council Leontief + lumpy economies with $\leq 2\%$ welfare gap vs. central MILP oracle. Publish all failure rates.	2–3 months 1.0 FTE OR/Econ \$0–\$100 compute (\$25k–\$45k grant)
Phase 1H: Low-Tech Human Probe (Trusted PostgreSQL + Web UI, Zero Crypto)	30–60 human participants in 3 WCs & 3 CCs completing a 3-Milestone planning cycle with explicit disclosure of trusted server.	KILL GATE 1H: $\geq 80\%$ completion in < 45 min/round; $\geq 30\%$ active override rate of solver defaults on perturbed prices; measure Amendment Gini alongside NASA-TLX.	1.5–2 months (Parallel w/ Ph. 1) 1.0 FTE UX/Dev \$50 VPS hosting
Phase 2: Distributed Coordination (Rust dpep-node, CBOR, QUIC, Wasmtime)	Multi-node WAN benchmarks (10^2 to 10^5 councils) vs. PostgreSQL baseline; fault-injection of censorship, equivocation & 20% partition.	KILL GATE 2: 100% detection of omitted/tampered proposals via signed receipts & challenge windows; < 30 s round latency at 10^5; overhead $\leq 5\times$ central DB.	3–4 months 2–3 FTEs \$1.5k–\$3k WAN cluster (\$150k–\$280k)
Phase 3: Integrated PQC, Prio3 & Pilot (FIPS 203–205, Prio3 VDAF, Mobile PWA)	Legacy Android hardware benchmarks; third-party crypto audit; parallel-economy field pilot with 10–30 cooperatives.	KILL GATE 3: Mandatory \$50k–\$150k external cryptographic audit before field use of custom Prio3 circuits; < 400 KB/round bandwidth & < 100 MB RAM on 5-yr-old Android.	6–12 months 4–8 FTEs + Audit \$250k–\$950k total programme

Table 9.2: Analytical Scaling Projections Across Five Population Scales (10-Round Annual Plan)

Metric (Category B Analytical Estimate)	100	1,000	10,000	100,000	1,000,000
Total Councils & Federations (M)	24	51	430	4,120	40,600
Priced Categories ($G = \|\Omega\|$)	30	80	500	2,000	10,000
Council Proposals / Round	20	40	400	4,000	40,000
Raw / Zstd Leaf Proposal Archive / Rd	104 KB	210 KB	2.1 MB	21 MB	208 MB (60 MB)
Mobile Client Bandwidth / Rd (Role-LP)	45–80 KB	80–130 KB	140–220 KB	190–300 KB	**250**–**400** KB
Layer-1 Checkpoint Growth / Yr (435 blks)	< 15 MB	< 25 MB	< 60 MB	< 120 MB	< **300** MB
Layer-2 Archive Growth / Yr (w/ indexes)	20–40 MB	0.15–0.3 GB	1.5–3 GB	15–30 GB	**120**–**220** GB
Mobile Client Peak RAM Target (Role-LP)	< 30 MB	< 40 MB	< 60 MB	< 80 MB	< **100** MB
Automated Compute & Net Time / Rd	< **0.2** s	**0.3**–**0.8** s	**1.5**–**4.0** s	**4.0**–**12.0** s	**9.0**–**27.0** s
Server Hosting Cost (Excl. Ops/Audits)	\$0–\$5/mo	\$10–\$25/mo	\$80–\$150/mo	\$400–\$800/mo	\$3,000–\$6,000/mo

CHAPTER 10

Open Theoretical, Institutional, Economic, and Technical Questions

> "A simulation incorporates more detail than a differential economic model, but still misses the richness of living interaction. As a result, the full characteristics of participatory economics and the diverse solutions to institutional problems associated with it will ultimately only reveal themselves in actual operations."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, p. 129[1]

We conclude our study by ranking the 15 most important unresolved questions across theoretical economics, institutional sociology, and distributed systems engineering, specifying how each will be investigated through either our simulation programme ([SIM]) or human-in-the-loop cooperative pilots ([HUMAN]):

Table 10.1: Ranked Research Agenda of Unresolved Questions (Priorities 1–15)

Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P1	Convergence Under Genuine Non-Convexities, Fixed Setup Costs & Leontief Complementarities	[ECON / OR] ([SIM])	Replace unconstrained Cobb-Douglas functions in pequod-cljs with mixed-integer setup costs and Leontief recipes; benchmark Algorithm 3 (ADMM proximal trust regions).
P2	Stability of Coupled Endogenous Income Feedback Across Rounds	[ECON] ([SIM])	Couple dynamic Society Account updates $(w_{\text{base}}^{(t)}, B_u^{(t)})$ into active rounds and map stability bounds of relaxation factor $\alpha_w \in (0, 1]$.
P3	Deliberative Complexity & Participation Fatigue in Multi-Round Planning	[INST / HCI] ([HUMAN])	Test the "3-Milestone Hybrid Schedule" (human rounds at $t = 1, 4, 8$ + automated intermediate solver rounds); measure NASA-TLX load.
P4	Biophysical Incommensurability vs. PDRM Pollution Pricing (O'Neill 2024)	[ECOLOGY] ([SIM/GOV])	Verify hard barrier convergence when aggregate demand hits absolute biophysical caps $(r^*, E_z^{\max})$ above local CAP PDRM pricing.
P5	Coarse-to-Granular SKU Disaggregation & Slack Buffer Sizing	[ECON / OR] ([SIM])	Simulate 50 coarse categories mapped to 2,000 SKUs under stochastic preference drift against slack $\eta \in [1\%, 10\%]$.
P6	Workplace Effort-Rating Dynamics: Solidarity vs. Surveillance or Parity	[INST] ([HUMAN])	Pilot with worker cooperatives comparing scaled effort multipliers vs. flat hourly parity ($\epsilon_u \equiv 1.0$) under BJC rotation.

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[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, p. 129.

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Rank	Unresolved Question	Domain	Experimental Investigation Protocol
P7	Preventing Industry Federation Cartelization & Gatekeeping (Kuch 2024)	[INST / ECON] ([HUMAN])	Test "Dual-Path Startup Accreditation" allowing rejected innovators to appeal to a sortition Innovation Jury + Consumer Federation.
P8	Calibrating Balanced Job Complexes Across Disparate Industries	[INST / OR] ([SIM/HUM])	Model cross-workplace BJC rotation as a bipartite matching problem in EFB accounting for travel & skill constraints.
P9	Inter-Regional Investment Apportionment Across Unequal Endowments	[ECON] ([SIM])	Compare per-capita vs. Porto Alegre need-weighted regional investment matrices over 10-year horizons.
P10	Reproductive & In-Home Caring Labour Integration (DEP Ch. 10)	[FEMINIST ECON] ([HUMAN])	Convene Women's and Caregivers' Caucuses to define care-allowance rules and household BJC norms.
P11	Byzantine Resilience Under Prolonged Regional Network Partitions	[DIST-SYS] ([SIM])	Benchmark Regional Proxy Extrapolation accuracy when 10% of councils are partitioned for multiple rounds.
P12	Standardisation of Compact Post-Quantum Threshold Signatures	[CRYPTO] ([SW-ARCH])	Maintain CryptoSuiteId agility to transition from multi-sig bitmaps to NIST threshold ML-DSA once standardised.
P13	Privacy Anonymity Sets in Very Small Rural Hamlets ($m < 20$)	[PRIVACY] ([SW-ARCH])	Merge pairwise-masking private consumption anonymity sets at the Ward level ($m \geq 250$) while keeping collective goods local.
P14	Endpoint Security vs. Open Modifiability on User Smartphones	[SECURITY] ([SW-ARCH])	Use local hardware keystore unlock + social anomaly verification without mandatory proprietary OS attestation lock-in.
P15	Bootstrapping a "Parallel Economy" Inside Surrounding Markets	[ADOPTION] ([HUMAN])	Implement a Dual-Price Fiat Reserve Bridge in Sandström's Society Account for federated cooperatives (PEPE Ch. 6 p. 129).

CHAPTER 11

Practical Adoption Strategy, Cooperative Pilots, and Theory of Transition

> "Conceivably one could run a 'shadow' participatory economy within an existing economy. This could occur in a subset of the whole economy, whether geographic or by industrial sector… not only [to] serve as a laboratory to test participatory planning, but also as a school, both educating the public and developing the means to implement participatory planning without undue disruption."
>
> — Michael Albert and Robin Hahnel, The Political Economy of Participatory Economics (1991), Ch. 6, "Experiment 2: Developing a Parallel Economy," pp. 129–130[1]

11.1 Distinguishing Research Simulation from Practical Cooperative Coordination

To avoid equating small-scale cooperative software with a full post-capitalist economy (R-POL-08), DPEP separates two distinct applications:

1. The Scientific Research Platform (Phases 1–2): A computational laboratory for testing whether multi-tier iterative planning algorithms converge under realistic industrial non-convexities, input-output networks, and strategic incentives.
2. The Practical Cooperative Coordination Suite (Phase 3): Modular local-first tools allowing existing worker cooperatives, consumer buying clubs, community land trusts (CLTs), and participatory budgeting municipalities to coordinate procurement, shared equipment, and inter-cooperative production within a surrounding capitalist market economy.

11.2 Theory of Transition: Market Boundaries, Voluntary Exit, and Legal Exposure (R-POL-08)

Any voluntary federation of cooperatives experimenting with participatory planning inside a surrounding market economy must navigate five structural realities:

1. Market-Priced Inputs, External Debts, and Taxes (The Dual-Price Boundary Constraint): Participating cooperatives cannot pay external steel suppliers, commercial landlords, bank mortgages, or national tax authorities in internal

[1] Albert and Hahnel, Political Economy of Participatory Economics, Ch. 6, pp. 129–130.

Table 11.1: Six Early Practical Applications and Their Institutional Requirements

Application	DPEP Modules Used	Viable Inside Market Economy?	Institutional / Ownership Changes Needed for Full Parecon Semantics
1. Cooperative Joint Procurement	Layer 2/3 demand aggregation across food co-ops & worker co-ops.	Yes (Immediately): Legal B2B purchasing cooperative.	Requires pooled working-capital reserves to buffer seasonal supplier price swings.
2. Shared Equipment & Tool Trusts	Stand-Alone Capital Asset ledger charging user-right fees (P_k).	Yes: Via a Cooperative Equipment Trust or CLT.	Member co-ops transfer title or long-term lease rights of heavy machinery/kitchens to the shared trust.
3. Community Mutual-Aid Distribution	Layer 3/4 Maxim-4 need allocation & non-transferable credits.	Yes: Non-commercial mutual-aid association.	Bounded by the volume of goods produced by volunteer or grant-funded community labour.
4. Inter-Cooperative Production Planning	Quarterly ADMM/tâtonnement clearing across 10–50 federated co-ops.	Partially (Dual-Price Regime): Needs Fiat Reserve Bridge.	External raw materials, rent, debt service, and taxes must still be settled in fiat currency (Eq. 11.1).
5. Municipal Participatory Budgeting	Layer 4 "What-If" explorer, Need Matrix & Prio3 voting.	Yes: Municipal public capital budgeting.	Vulnerable to municipal coalition reversals and fiscal austerity (as in Porto Alegre's 2017 suspension).
6. Regional Co-op Supply Networks	Multi-year investment (σ_{inv}) & EFB worker reallocation.	Partially (e.g., Mondragón / Cecosesola).	Full Parecon requires abolishing enterprise profit retention and equalising BJC empowerment across plants.

effort credits. Every pilot federation must enforce an explicit External Fiat Clearing Constraint alongside its internal Society Account:

$$\sum_{j \in \mathcal{J}} P_{\text{ext}}^{\text{fiat}} \cdot \bar{a}_j^{\text{export}} + D_{\text{fiat}}^{\text{dues}} \geq \sum_{j \in \mathcal{J}} P_{\text{ext}}^{\text{fiat}} \cdot \underline{a}_j^{\text{import}} + F^{\text{taxes}} + F^{\text{debt}} \tag{11.1}$$

where external imports $\underline{a}_j^{\text{import}}$ carry an internal shadow price $\lambda_{\text{fiat}} P_{\text{ext}}^{\text{fiat}}$ reflecting the scarcity of external fiat revenue.

2. Strict Voluntarism, Right of Exit, and Non-Participants (Learning from Aragon 1936): Informed by the conflict between Aragonese collectives and "individualist" smallholders (Casanova 2005), membership in any DPEP pilot or transitional federation must be strictly voluntary with a constitutionally codified Right of Exit: exiting individuals or cooperatives settle their net savings credits and refundable capital shares according to pre-agreed cooperative bylaws without confiscation, and non-participating artisans/households trade with the federation across transparent external interfaces.

3. Graduated Sanctions Without Coercive Labour Discipline: Following Ostrom (1990), failure to meet council commitments triggers: (a) automatic SB_j / SC_j accounting adjustment, (b) UFB peer mediation, and (c) for persistent bad-faith breach, revocation of federation accreditation and access to shared federation inputs/credit—never forced labour.

4. Jurisdiction-Specific Legal Exposure (Informational Analysis, Not Legal Advice):

   - Competition / Antitrust Law (US Sherman Act §1; EU TFEU Art. 101; UK Competition Act 1998): Independent corporations that coordinate prices and output quotas through an IFB risk prosecution as a horizontal price-fixing cartel unless structured under statutory cooperative/agricultural exemptions (e.g., US Capper-Volstead Act, EU producer-organisation rules) or incorporated as operating divisions of a single secondary cooperative / common-ownership consortium.
   - Taxation of Multilateral Barter & Clearing (US IRS Form 1099-B; EU-/UK VAT): Multilateral credit clearing is generally taxable at fair market value; pilot deployments must generate statutory fiat-equivalent tax reports where required by domestic law.
   - Data Protection & Right to Erasure (EU/UK GDPR Art. 17): DPEP avoids immutable personal ledger data by keeping personal profiles local, using ephemeral Prio3 secret shares, and storing qualitative Lists behind mutable council pointers with cryptographic tombstone erasure (R-TECH-04).

CHAPTER 12

Six-Discipline Critical Examination and Remaining Weaknesses

Table 12.1 evaluates how the DPEP architecture withstands critical scrutiny across six disciplines, identifying the three strongest remaining criticisms in each field and their current resolution status:

Table 12.1: Six-Discipline Critical Examination: Strongest Remaining Criticisms and Resolution Status

Discipline	Three Strongest Remaining Criticisms	Status (Resolved / Mitigated / Open) & Explanation
1. Participatory Economics Researcher	1a. SB_j/SC_j effort caps risk penalising workers in older plants (R-ECON-09). 1b. Tier-1 Federation MILP for lumpy plants could centralise authority (Kuch 2024). 1c. 3-Milestone schedule reduces inter-round social learning (R-POL-03).	1a: Mitigated via $P_k k_j^0$ rental calibration + Kornai-capped transition buffers. 1b: Open Trade-Off — indivisible plants cannot clear via marginal price steps alone (R-ECON-04); MILP choices require council ratification. 1c: Open — tested empirically in Phase 1H.
2. Operations Researcher	2a. No price/ADMM method guarantees global optimum on general non-convex MIPs. 2b. ADMM can converge slowly if $(I - A)$ is ill-conditioned. 2c. Coarse-to-SKU disaggregation fails if within-class MRTs diverge (R-ECON-11).	2a: Mitigated via Two-Tier MILP + Convex ADMM; Open empirically until Phase 1 measures the MIP duality gap. 2b: Open — tested in Phase 1 (preconditioning / Anderson acceleration). 2c: Open — tested in Phase 1 Experiment 5/6.
3. Distributed Systems Engineer	3a. Prolonged regional network partitions force proxy approximations (P11). 3b. Full leaf DA (60 MB/rd) scales linearly with $M \cdot s$ above $N = 10^7$. 3c. Offline depot credit allows bounded double-spending if device cloned (R-TECH-04).	3a: Mitigated by explicit Active/Partitioned bitmaps (R-SEC-06) & slack SL^E. 3b: Mitigated up to $N = 10^7$ via regional sharding; requires Data Availability Sampling at global scale. 3c: Mitigated (bounded by daily allowance cap, reconciled at Ward rollup).
4. Cryptographer	4a. Prio3 VDAF relies on 1-of-3 non-collusion among council/ward/fed aggregators. 4b. Post-quantum threshold sigs & PQ-VRFs are not yet NIST-finalised. 4c. Custom Prio3 inner-product budget circuit is unbenchmarked & unaudited.	4a: Mitigated by institutional separation of the 3 aggregators (ACORN retained as zero-server backup). 4b: Mitigated via commit-reveal sortition & multi-sig bitmaps. 4c: Open — gated by Kill Gate 3 ($50k–$150k external audit).
5. Cooperative Practitioner	5a. Meeting fatigue & deference to vocal members (R-POL-03, R-SRC-06). 5b. Peer effort ratings rejected by many co-ops in favour of flat parity (P6). 5c. Market-priced inputs, taxes & antitrust exposure in pilots (R-POL-08).	5a: Open — primary subject of Phase 1H (measuring Amendment Gini & override rates). 5b: Resolved by supporting flat hourly parity ($\epsilon_u \equiv 1.0$) natively. 5c: Mitigated via Ch. 11 Fiat Reserve Bridge & secondary-co-op legal structure.
6. Sceptical Economist	6a. Strategic capacity understatement & Hurwicz (1972) impossibility (R-ECON-08). 6b. Local CAP bilateral monopoly over emission permits (R-ECON-08). 6c. Dynamic innovation risk-taking without entrepreneurial profit (R-POL-04).	6a: Mitigated (vs. central planning) by public blueprints & non-ratcheted endowment fees, but Open as a fundamental limit. 6b: Mitigated by $E_z^{\max}$ ceilings & Environmental Courts. 6c: Mitigated via Dual-Path Startup Accreditation, Open empirically.

CHAPTER A

Scientific Claims, Evidence, and Validation Status Register

Table A.1: Master Register of Scientific and Engineering Claims

Claim ID	Claim & Assumptions	Class	Evidence & Limitations	Validation & Status
CLM-ECON-01	Continuous-time FMPE gradient convergence under strict convexity & gross substitutability.	Established	Arrow-Hurwicz-Uzawa (1958); Heal (1973); PEPE Ch. 5. Fails on non-convex/Scarf sets.	Phase 1 Exp. 1. Verified
CLM-ECON-02	pequod-cljs Cobb-Douglas convergence in 11.85 rds (Y1, 5%), 19.20 rds (3%), 6.53 rds (Y2).	Source-supported	Audited across dep1ex61..100.csv (8bd46b5c). Fixed nominal income; 2.5s timing has no log.	Phase 1 Exp. 1. Verified (CSV); timing Unverified
CLM-ECON-03	Naive multiplicative price tâtonnement + trust regions cannot guarantee non-convex/Leontief convergence.	Established (Negative)	Demonstrated by verify_economics.py: Scarf cycle stays $\geq 10.5\%$; single plant stays $\geq 50.4\%$.	Addressed by Two-Tier MILP + ADMM. Verified
CLM-ECON-04	pequod-cljs create-toothache makes $\sum b_m + c/k \in [1.07, 1.67] > 1$ across 100% of 150k WCs (unbounded / saddle point).	Established / Code	Proven in Theorem 2.1 & audited across all 70 dep1ex*.clj files (R-ECON-06).	Phase 1 Exp. 3. Verified
CLM-ECON-05	Shapley–Folkman smoothing across $K \geq 20$ heterogeneous plants shrinks non-clearing gap below 0.5%.	Established / Analytical	Starr (1969); verified in verify_economics.py ($K = 20 \implies \rho = 0.40\%$). Large $K = 1$ plants still need MILP.	Phase 1 Exp. 3. Verified
CLM-ECON-06	Numéraire $w_{\text{base}} \equiv 1.0$ & effort-weighted hours $H^{\text{eff}} = \sum h_u \epsilon_u$ close macro accounts and eliminate scale drift.	Established	Walras's Law; verified in verify_economics.py (R-ECON-05, R-ECON-07).	Phase 1 Exp. 5. Verified
CLM-TECH-01	Automated federated round aggregation at $N = 10^6$ completes in 9–27 s across 50 servers.	Analytical / Target	Derived from 60 MB compressed leaf archive + 616 KB block (1.5–3× factor). Human rounds take days.	Phase 2 Kill Gate 2. Unverified

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Claim ID	Claim & Assumptions	Class	Evidence & Limitations	Validation & Status
CLM-TECH-02	Mobile client (Role-LP) round bandwidth is 250–400 KB/round and peak RAM < 100 MB.	Analytical / Target	Recomputed in verify_security_and_scoring.py (R-TECH-02: 145 KB sigs + 75 KB Prio3 + deltas).	Phase 3 Kill Gate 3. Partially verified (Bytes); RAM Unverified
CLM-SEC-01	Prio3 VDAF secret sharing ($K = 3$) + FLP binds private household budget compliance to counted demand with offline dropout resilience.	Source-supported / Analytical	IETF draft-irtf-cfrg-vdaf-16; Corrigan-Gibbs & Boneh (2017). Requires 1-of-3 honest aggregator.	Phase 3 External Audit. Partially verified
CLM-SEC-02	Two-Root BLAKE3 Merkle tree (512 B proof) + full leaf DA (60 MB) + Active/Partitioned bitmap detects tampering & censorship.	Analytical	Remediates 1.28 MB naive MST proof bloat (R-SEC-06). Requires ≥ 1 honest relay & DA auditor.	Phase 2 Fault-Injection. Partially verified
CLM-SEC-03	Hybrid X25519+ML-KEM-768, Ed25519+ML-DSA-65, and SLH-DSA-192s provide uniform NIST Level-3 PQC security.	Established	NIST FIPS 203, 204, 205 (2024) (R-SEC-10). Auxiliary threshold/VRF primitives remain draft.	Phase 3 Crypto Review. Verified (Specs)
CLM-SEC-04	INV-4B blocks digital hours inflation, but physical output, emissions & carousel fraud require physical euthynai audits.	Analytical	Boundary analysis (R-SEC-07, R-SEC-11). Cryptography cannot prove physical reality.	Phase 3 Pilot Rules. Verified (Boundary)
CLM-POL-01	Active-choice UI (≥ 3 Pareto options) prevents local QP solver defaults from becoming unaccountable decisions.	Experimental target	Addresses R-POL-03 & Dapprich (2024). Human override rate is an empirical question.	Phase 1H Kill Gate 1H. Unverified

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